

Smithsonian Institution

National Museum of Natural History

10th STREET AND CONSTITUTION AVENUE | WASHINGTON DC | DC 20560

Upgrade Sanitary and Storm Systems

Bid Set Issuance

Project Manual



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1.0 EXECUTIVE SUMMARY

1.1 PROJECT EXECUTIVE SUMMARY

The National Museum of Natural History has existed on the National Mall for more than a century. In that time, the building grew significantly from the original footprint to meet the ever-changing needs of the institution. Originally opened in 1910, the building was expanded in the 1960s with the addition of the East and West Wings, which provided storage space for the museums' vast collections as well as supporting staff spaces. Over the next several decades the original building's courtyards were infilled to provide additional necessary square footage for both public and private needs. In addition, the Chiller Plant was constructed in the east parking lot to better support building-wide infrastructure needs.

Many decades of use have left the Storm and Sanitary Systems in need of extensive repair and in many cases full replacement in various portions of the building. As problems have developed over time, the museum has repaired leaks as best they can, but the number and frequency of repairs are becoming unsustainable. This growing problem has placed numerous collections, research, exhibit, and office spaces at a level of unacceptable risk. A significant intervention is needed to address these failing systems so that the museum can continue its mission.

As part of a comprehensive Implementation Study, the Design Team surveyed the Storm and Sanitary Systems throughout the building to gain a comprehensive understanding of the condition. The exhaustive survey effort uncovered many issues with the Storm and Sanitary piping systems. In addition to systems needing replacement because of being at the end of their useful life, many components were found to be inadequately sized, improperly installed or contained materials not in alignment with current Smithsonian Design Standards (SDS). Based on the findings, a series of options were developed to address the issues and compiled into the Implementation Study Report. The report made recommendations for each issue and weighed the pros and cons of each. Since the report has been issued, SI has made decisions on how to proceed. Those decisions are reflected in this 100% Construction Documents Submission.

The path forward for the Storm and Sanitary Systems are highlighted below:

- **East and West Wing Toilet Rooms**

The renovation of a stack of restrooms one at a time will be phased, starting from the top and working down. To facilitate this a new stack will be installed prior to commencement of demolition to allow for a more efficient construction process.

- **East and West Wing Storm Piping**

A single storm piping stack will be replaced in kind one stack at a time. Roof drains will be plugged while the piping in a stack is methodically replaced floor by floor. New Roof drains will not be provided.

- **East and West Wing Office Sinks**

Existing office sinks will be maintained in the East and West Wings. The sanitary piping associated with these existing sinks will be replaced. Existing piping will be demolished and abandoned in place. Domestic water to these sinks will not be impacted.

1.0 EXECUTIVE SUMMARY

- **Main Building Storm Piping**

New stacks will be provided, from the attic to the tunnels that are offset to peripheral spaces that have potential to have less impact on building operations, both public and private. Existing stacks will be abandoned in place to avoid disruptive and costly demolition.

- **Main Building Roof Scuppers**

The roof scuppers connected to the Main Building's stormwater system feature a variety of attachment methods to the existing copper roofing. Internal repairs will be conducted on the scuppers at locations where leaks have been identified.

- **Main Building Third Floor Office Sinks**

Remove office sinks, cap, and abandon associated piping in place. Kitchenettes will be provided to give users access to sinks in this area of the building.

- **Main Building Toilet Rooms**

Main Building toilet rooms will be renovated. New offset stacks will be provided to facilitate the renovation of restrooms.

- **Tunnel Piping Replacement**

Much of the existing piping in the Main Building requires replacement. New, larger tunnel access points throughout the Ground Floor of the Main Building will be needed to better support the replacement effort.

- **Exterior Storm Piping Replacement**

At seven locations, the storm piping increases in size as it exits the building. This change results from the consolidation of stormwater piping within the Main Building, East Wing, and West Wing. As internal piping systems merge, the combined flow necessitates larger pipe diameters downstream. Consequently, exterior excavation is required to accommodate the upsized piping and to reconnect it to the nearest existing storm drain structures. Additional details regarding this scope of work are provided in the Plumbing and Civil Engineering sections of this report.

2.0 PROGRAM SUMMARY

The Upgrade Sanitary and Storm Project maintains the existing restrooms throughout the building while upgrading the necessary infrastructure that supports them. Modifications throughout the building will improve the accessibility of the space and provide spaces for gender inclusiveness. The core program function of these spaces remains unchanged.

2.1 TABLE OF SPACE TYPES

2.1.1 PUBLIC RESTROOMS

Restrooms serving the public in the Wings and Main Building will be revitalized as part of this project. Each space will receive all new finishes and improve layouts where applicable. To the greatest extent possible, existing fixture counts will be maintained.

2.1.2 NON-PUBLIC RESTROOMS

Restrooms in staff areas, particularly in the Wings, will be revamped. Many of these spaces will have adjustments to the layout that improve the overall flow. Additionally, underutilized space previously used as lounges in women's restrooms will be utilized to add additional useable space to restroom facilities.

2.1.3 FAMILY RESTROOMS

Family Restrooms are an amenity that provides a gender neutral, universally accessible restroom for both the public and non-public. The key difference in this restroom from a single-use gender neutral room is the adult changing table, which provides an amenity previously not provided in the building.

2.1.4 PANTRIES

To minimize the impacts of sanitary pipe replacement in the Main Building, office sinks are being removed on the Third Floor. To allow staff access to sinks for personal use, two pantries are being provided on the Third Floor of the Main Building. Spaces selected for this function are identified below.

2.1.5 THIRD FLOOR - ROSE ROOM

The Rose Room is a large conference room that is used for lectures, conferences, and meetings. The pantry will replace the existing sink in this space. The purpose of this amenity is to serve the staff in the East Range, providing them with a place to store and prepare their lunch or cater for an event happening in the room.

2.1.6 THIRD FLOOR SHARED OFFICE (ROOM 317)

The Shared Office in the East Hall contains a space outside of the room to be utilized as a pantry. There will be some space taken out of the adjacent custodial closet to house this pantry. The purpose of improving this amenity is to serve the staff in the East Hall, providing them with a place to store and prepare their lunch.

2.0 PROGRAM SUMMARY

2.1.7 Wing Dash Elevator Lobbies

The elevator lobbies outside of the wing restrooms will be heavily trafficked and will take constant wear and tear as part of the renovations of the restrooms. These areas will be improved through new finishes and new water fountains, to provide an improved transitional space into the new restrooms for both the public and non-public.

2.1.8 Wing Dash Labs

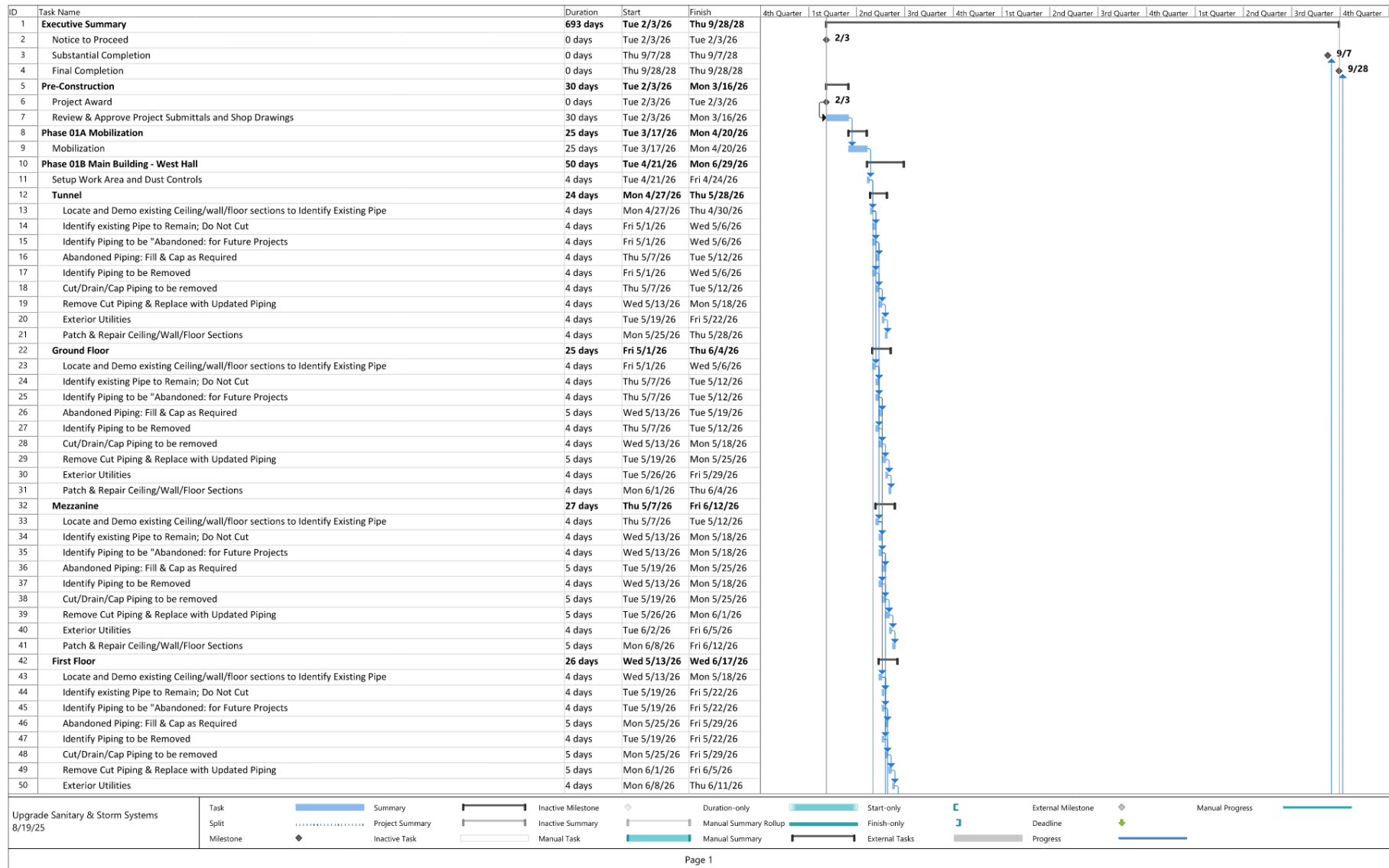
The labs adjacent to wing dash restrooms on the first and third floor of both wings will be utilized as part of the sanitary piping scope. These rooms will be used to access sanitary pipes as they come from the restrooms above. Where labs have outdated ceilings and fixtures, they will receive new ceilings, lighting, diffusers, sprinklers, and paint. Where labs have been recently renovated, they will receive new paint.

3.0 SCHEDULE

3.1 CONSTRUCTION SCHEDULE

A construction schedule for the implementation of this project is included in this section. This schedule will be updated throughout the design. Future phases of the project will identify the preferred sequencing of the work through the building.





ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
51	Patch & Repair Ceiling/Wall/Floor Sections	4 days	Fri 6/12/26	Wed 6/17/26									
52	Second Floor	26 days	Tue 5/19/26	Tue 6/23/26									
53	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	4 days	Tue 5/19/26	Fri 5/22/26									
54	Identify existing Pipe to Remain; Do Not Cut	4 days	Mon 5/25/26	Thu 5/28/26									
55	Identify Piping to be "Abandoned: for Future Projects	4 days	Mon 5/25/26	Thu 5/28/26									
56	Abandoned Piping: Fill & Cap as Required	5 days	Fri 5/29/26	Thu 6/4/26									
57	Identify Piping to be Removed	4 days	Mon 5/25/26	Thu 5/28/26									
58	Cut/Drain/Cap Piping to be removed	5 days	Fri 5/29/26	Thu 6/4/26									
59	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 6/5/26	Thu 6/11/26									
60	Exterior Utilities	4 days	Fri 6/12/26	Wed 6/17/26									
61	Patch & Repair Ceiling/Wall/Floor Sections	4 days	Thu 6/18/26	Tue 6/23/26									
62	Third Floor	23 days	Mon 5/25/26	Wed 6/24/26									
63	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 5/25/26	Wed 5/27/26									
64	Identify existing Pipe to Remain; Do Not Cut	3 days	Thu 5/28/26	Mon 6/1/26									
65	Identify Piping to be "Abandoned: for Future Projects	4 days	Fri 5/29/26	Wed 6/3/26									
66	Abandoned Piping: Fill & Cap as Required	4 days	Thu 6/4/26	Tue 6/9/26									
67	Identify Piping to be Removed	4 days	Thu 5/28/26	Tue 6/2/26									
68	Cut/Drain/Cap Piping to be removed	4 days	Wed 6/3/26	Mon 6/8/26									
69	Remove Cut Piping & Replace with Updated Piping	4 days	Tue 6/9/26	Fri 6/12/26									
70	Exterior Utilities	4 days	Mon 6/15/26	Thu 6/18/26									
71	Patch & Repair Ceiling/Wall/Floor Sections	4 days	Fri 6/19/26	Wed 6/24/26									
72	Attic	23 days	Thu 5/28/26	Mon 6/29/26									
73	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 5/28/26	Mon 6/1/26									
74	Identify existing Pipe to Remain; Do Not Cut	4 days	Tue 6/2/26	Fri 6/5/26									
75	Identify Piping to be "Abandoned: for Future Projects	4 days	Thu 6/4/26	Tue 6/9/26									
76	Abandoned Piping: Fill & Cap as Required	4 days	Wed 6/10/26	Mon 6/15/26									
77	Identify Piping to be Removed	4 days	Tue 6/2/26	Fri 6/5/26									
78	Cut/Drain/Cap Piping to be removed	4 days	Mon 6/8/26	Thu 6/11/26									
79	Remove Cut Piping & Replace with Updated Piping	4 days	Fri 6/12/26	Wed 6/17/26									
80	Exterior Utilities	4 days	Thu 6/18/26	Tue 6/23/26									
81	Patch & Repair Ceiling/Wall/Floor Sections	4 days	Wed 6/24/26	Mon 6/29/26									
82	Phase 01C Main Building - Rotunda	50 days	Tue 6/30/26	Mon 9/7/26									
83	Setup Work Area and Dust Controls	2 days	Tue 6/30/26	Wed 7/1/26									
84	Tunnel	28 days	Thu 7/2/26	Mon 8/10/26									
85	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 7/2/26	Mon 7/6/26									
86	Identify existing Pipe to Remain; Do Not Cut	3 days	Tue 7/7/26	Thu 7/9/26									
87	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 7/7/26	Mon 7/13/26									
88	Abandoned Piping: Fill & Cap as Required	5 days	Tue 7/14/26	Mon 7/20/26									
89	Identify Piping to be Removed	5 days	Tue 7/7/26	Mon 7/13/26									
90	Cut/Drain/Cap Piping to be removed	5 days	Tue 7/14/26	Mon 7/20/26									
91	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 7/21/26	Mon 7/27/26									
92	Exterior Utilities	5 days	Tue 7/28/26	Mon 8/3/26									
93	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 8/4/26	Mon 8/10/26									
94	Ground Floor	28 days	Tue 7/7/26	Thu 8/13/26									
95	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 7/7/26	Thu 7/9/26									
96	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 7/10/26	Thu 7/16/26									
97	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 7/14/26	Mon 7/20/26									
98	Abandoned Piping: Fill & Cap as Required	5 days	Tue 7/21/26	Mon 7/27/26									
99	Identify Piping to be Removed	5 days	Fri 7/10/26	Thu 7/16/26									
100	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/17/26	Thu 7/23/26									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

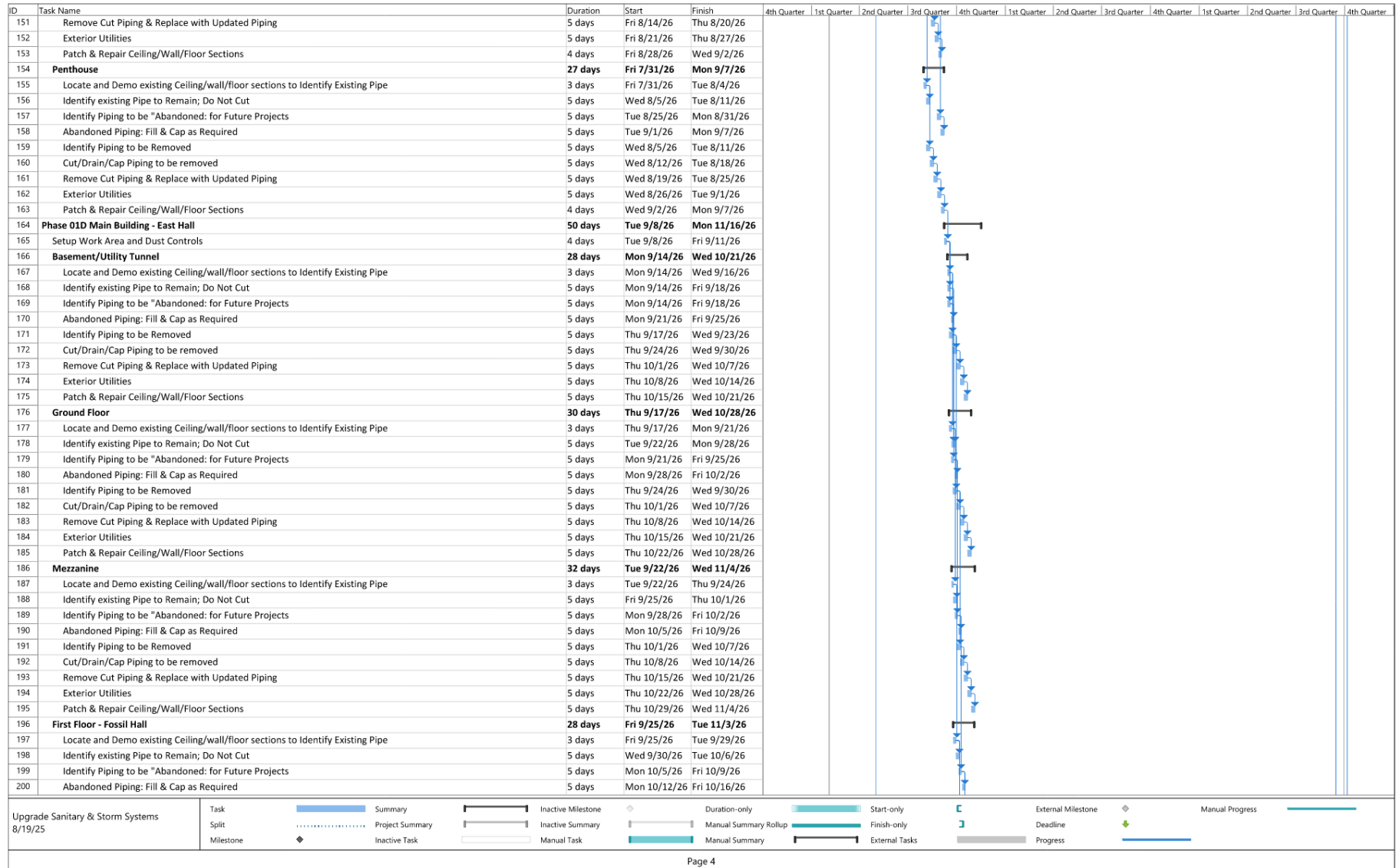
Deadline

Progress

Manual Progress

Page 2

ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
101	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/24/26	Thu 7/30/26													
102	Exterior Utilities	5 days	Fri 7/31/26	Thu 8/6/26													
103	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 8/7/26	Thu 8/13/26													
104	Mezzanine	28 days	Fri 7/10/26	Tue 8/18/26													
105	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Fri 7/10/26	Tue 7/14/26													
106	Identify existing Pipe to Remain; Do Not Cut	5 days	Wed 7/15/26	Tue 7/21/26													
107	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 7/21/26	Mon 7/27/26													
108	Abandoned Piping: Fill & Cap as Required	5 days	Tue 7/28/26	Mon 8/3/26													
109	Identify Piping to be Removed	5 days	Wed 7/15/26	Tue 7/21/26													
110	Cut/Drain/Cap Piping to be removed	5 days	Wed 7/22/26	Tue 7/28/26													
111	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 7/29/26	Tue 8/4/26													
112	Exterior Utilities	5 days	Wed 8/5/26	Tue 8/11/26													
113	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 8/12/26	Tue 8/18/26													
114	First Floor	28 days	Wed 7/15/26	Fri 8/21/26													
115	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 7/15/26	Fri 7/17/26													
116	Identify existing Pipe to Remain; Do Not Cut	5 days	Mon 7/20/26	Fri 7/24/26													
117	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 7/28/26	Mon 8/3/26													
118	Abandoned Piping: Fill & Cap as Required	5 days	Tue 8/4/26	Mon 8/10/26													
119	Identify Piping to be Removed	5 days	Mon 7/20/26	Fri 7/24/26													
120	Cut/Drain/Cap Piping to be removed	5 days	Mon 7/27/26	Fri 7/31/26													
121	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 8/3/26	Fri 8/7/26													
122	Exterior Utilities	5 days	Mon 8/10/26	Fri 8/14/26													
123	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 8/17/26	Fri 8/21/26													
124	Second Floor	28 days	Mon 7/20/26	Wed 8/26/26													
125	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 7/20/26	Wed 7/22/26													
126	Identify existing Pipe to Remain; Do Not Cut	5 days	Thu 7/23/26	Wed 7/29/26													
127	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 8/4/26	Mon 8/10/26													
128	Abandoned Piping: Fill & Cap as Required	5 days	Tue 8/11/26	Mon 8/17/26													
129	Identify Piping to be Removed	5 days	Thu 7/23/26	Wed 7/29/26													
130	Cut/Drain/Cap Piping to be removed	5 days	Thu 7/30/26	Wed 8/5/26													
131	Remove Cut Piping & Replace with Updated Piping	5 days	Thu 8/6/26	Wed 8/12/26													
132	Exterior Utilities	5 days	Thu 8/13/26	Wed 8/19/26													
133	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 8/20/26	Wed 8/26/26													
134	Third Floor	28 days	Thu 7/23/26	Mon 8/31/26													
135	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 7/23/26	Mon 7/27/26													
136	Identify existing Pipe to Remain; Do Not Cut	5 days	Tue 7/28/26	Mon 8/3/26													
137	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 8/11/26	Mon 8/17/26													
138	Abandoned Piping: Fill & Cap as Required	5 days	Tue 8/18/26	Mon 8/24/26													
139	Identify Piping to be Removed	5 days	Tue 7/28/26	Mon 8/3/26													
140	Cut/Drain/Cap Piping to be removed	5 days	Tue 8/4/26	Mon 8/10/26													
141	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 8/11/26	Mon 8/17/26													
142	Exterior Utilities	5 days	Tue 8/18/26	Mon 8/24/26													
143	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 8/25/26	Mon 8/31/26													
144	Attic	27 days	Tue 7/28/26	Wed 9/2/26													
145	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 7/28/26	Thu 7/30/26													
146	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 7/31/26	Thu 8/6/26													
147	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 8/18/26	Mon 8/24/26													
148	Abandoned Piping: Fill & Cap as Required	5 days	Tue 8/25/26	Mon 8/31/26													
149	Identify Piping to be Removed	5 days	Fri 7/31/26	Thu 8/6/26													
150	Cut/Drain/Cap Piping to be removed	5 days	Fri 8/7/26	Thu 8/13/26													
Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress																	
Upgrade Sanitary & Storm Systems 8/19/25																	



ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
201	Identify Piping to be Removed	5 days	Wed 9/30/26	Tue 10/6/26									
202	Cut/Drain/Cap Piping to be removed	5 days	Wed 10/7/26	Tue 10/13/26									
203	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 10/14/26	Tue 10/20/26									
204	Exterior Utilities	5 days	Wed 10/21/26	Tue 10/27/26									
205	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 10/28/26	Tue 11/3/26									
206	Second Floor - Geology	28 days	Wed 9/30/26	Fri 11/6/26									
207	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 9/30/26	Fri 10/2/26									
208	Identify existing Pipe to Remain; Do Not Cut	5 days	Mon 10/5/26	Fri 10/9/26									
209	Identify Piping to be "Abandoned: for Future Projects	5 days	Mon 10/12/26	Fri 10/16/26									
210	Abandoned Piping: Fill & Cap as Required	5 days	Mon 10/19/26	Fri 10/23/26									
211	Identify Piping to be Removed	5 days	Mon 10/5/26	Fri 10/9/26									
212	Cut/Drain/Cap Piping to be removed	5 days	Mon 10/12/26	Fri 10/16/26									
213	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 10/19/26	Fri 10/23/26									
214	Exterior Utilities	5 days	Mon 10/26/26	Fri 10/30/26									
215	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 11/2/26	Fri 11/6/26									
216	Third Floor	28 days	Mon 10/5/26	Wed 11/11/26									
217	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 10/5/26	Wed 10/7/26									
218	Identify existing Pipe to Remain; Do Not Cut	5 days	Thu 10/8/26	Wed 10/14/26									
219	Identify Piping to be "Abandoned: for Future Projects	5 days	Mon 10/19/26	Fri 10/23/26									
220	Abandoned Piping: Fill & Cap as Required	5 days	Mon 10/26/26	Fri 10/30/26									
221	Identify Piping to be Removed	5 days	Thu 10/8/26	Wed 10/14/26									
222	Cut/Drain/Cap Piping to be removed	5 days	Thu 10/15/26	Wed 10/21/26									
223	Remove Cut Piping & Replace with Updated Piping	5 days	Thu 10/22/26	Wed 10/28/26									
224	Exterior Utilities	5 days	Thu 10/29/26	Wed 11/4/26									
225	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 11/5/26	Wed 11/11/26									
226	Attic	28 days	Thu 10/8/26	Mon 11/16/26									
227	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 10/8/26	Mon 10/12/26									
228	Identify existing Pipe to Remain; Do Not Cut	5 days	Tue 10/13/26	Mon 10/19/26									
229	Identify Piping to be "Abandoned: for Future Projects	5 days	Mon 10/26/26	Fri 10/30/26									
230	Abandoned Piping: Fill & Cap as Required	5 days	Mon 11/2/26	Fri 11/6/26									
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232	Cut/Drain/Cap Piping to be removed	5 days	Tue 10/20/26	Mon 10/26/26									
233	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 10/27/26	Mon 11/2/26									
234	Exterior Utilities	5 days	Tue 11/3/26	Mon 11/9/26									
235	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 11/10/26	Mon 11/16/26									
236	Roof	3 days	Tue 11/10/26	Thu 11/12/26									
237	Make Connections to Current Roof System	3 days	Tue 11/10/26	Thu 11/12/26									
238	Phase 01E Main Building - North Hall	75 days	Fri 11/13/26	Thu 2/25/27									
239	Setup Work Area and Dust Controls	4 days	Fri 11/13/26	Wed 11/18/26									
240	Tunnel	31 days	Thu 11/19/26	Thu 12/31/26									
241	North	28 days	Thu 11/19/26	Mon 12/28/26									
242	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 11/19/26	Mon 11/23/26									
243	Identify existing Pipe to Remain; Do Not Cut	5 days	Tue 11/24/26	Mon 11/30/26									
244	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 11/19/26	Wed 11/25/26									
245	Abandoned Piping: Fill & Cap as Required	5 days	Thu 11/26/26	Wed 12/2/26									
246	Identify Piping to be Removed	5 days	Tue 11/24/26	Mon 11/30/26									
247	Cut/Drain/Cap Piping to be removed	5 days	Tue 12/1/26	Mon 12/7/26									
248	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 12/8/26	Mon 12/14/26									
249	Exterior Utilities	5 days	Tue 12/15/26	Mon 12/21/26									
250	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 12/22/26	Mon 12/28/26									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
251	South	28 days	Tue 11/24/26	Thu 12/31/26									
252	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 11/24/26	Thu 11/26/26									
253	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 11/27/26	Thu 12/3/26									
254	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 11/26/26	Wed 12/2/26									
255	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/3/26	Wed 12/9/26									
256	Identify Piping to be Removed	5 days	Fri 11/27/26	Thu 12/3/26									
257	Cut/Drain/Cap Piping to be removed	5 days	Fri 12/4/26	Thu 12/10/26									
258	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 12/11/26	Thu 12/17/26									
259	Exterior Utilities	5 days	Fri 12/18/26	Thu 12/24/26									
260	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 12/25/26	Thu 12/31/26									
261	Ground Floor	31 days	Fri 11/27/26	Fri 1/8/27									
262	North	28 days	Fri 11/27/26	Tue 1/5/27									
263	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Fri 11/27/26	Tue 12/1/26									
264	Identify existing Pipe to Remain; Do Not Cut	5 days	Wed 12/2/26	Tue 12/8/26									
265	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/3/26	Wed 12/9/26									
266	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/10/26	Wed 12/16/26									
267	Identify Piping to be Removed	5 days	Wed 12/2/26	Tue 12/8/26									
268	Cut/Drain/Cap Piping to be removed	5 days	Wed 12/9/26	Tue 12/15/26									
269	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 12/16/26	Tue 12/22/26									
270	Exterior Utilities	5 days	Wed 12/23/26	Tue 12/29/26									
271	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 12/30/26	Tue 1/5/27									
272	South	28 days	Wed 12/2/26	Fri 1/8/27									
273	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 12/2/26	Fri 12/4/26									
274	Identify existing Pipe to Remain; Do Not Cut	5 days	Mon 12/7/26	Fri 12/11/26									
275	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/10/26	Wed 12/16/26									
276	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/17/26	Wed 12/23/26									
277	Identify Piping to be Removed	5 days	Mon 12/7/26	Fri 12/11/26									
278	Cut/Drain/Cap Piping to be removed	5 days	Mon 12/14/26	Fri 12/18/26									
279	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 12/21/26	Fri 12/25/26									
280	Exterior Utilities	5 days	Mon 12/28/26	Fri 1/1/27									
281	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 1/4/27	Fri 1/8/27									
282	First Floor	31 days	Mon 12/7/26	Mon 1/18/27									
283	North	28 days	Mon 12/7/26	Wed 1/13/27									
284	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 12/7/26	Wed 12/9/26									
285	Identify existing Pipe to Remain; Do Not Cut	5 days	Thu 12/10/26	Wed 12/16/26									
286	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/17/26	Wed 12/23/26									
287	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/24/26	Wed 12/30/26									
288	Identify Piping to be Removed	5 days	Thu 12/10/26	Wed 12/16/26									
289	Cut/Drain/Cap Piping to be removed	5 days	Thu 12/17/26	Wed 12/23/26									
290	Remove Cut Piping & Replace with Updated Piping	5 days	Thu 12/24/26	Wed 12/30/26									
291	Make new Connection to Exterior Systems	5 days	Thu 12/31/26	Wed 1/6/27									
292	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 1/7/27	Wed 1/13/27									
293	South	28 days	Thu 12/10/26	Mon 1/18/27									
294	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 12/10/26	Mon 12/14/26									
295	Identify existing Pipe to Remain; Do Not Cut	5 days	Tue 12/15/26	Mon 12/21/26									
296	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/24/26	Wed 12/30/26									
297	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/31/26	Wed 1/6/27									
298	Identify Piping to be Removed	5 days	Tue 12/15/26	Mon 12/21/26									
299	Cut/Drain/Cap Piping to be removed	5 days	Tue 12/22/26	Mon 12/28/26									
300	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 12/29/26	Mon 1/4/27									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
301	Make new Connection to Exterior Systems	5 days	Tue 1/5/27	Mon 1/11/27									
302	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 1/12/27	Mon 1/18/27									
303	Second Floor	33 days	Tue 12/15/26	Thu 1/28/27									
304	North	28 days	Tue 12/15/26	Thu 1/21/27									
305	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 12/15/26	Thu 12/17/26									
306	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 12/18/26	Thu 12/24/26									
307	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/31/26	Wed 1/6/27									
308	Abandoned Piping: Fill & Cap as Required	5 days	Thu 1/7/27	Wed 1/13/27									
309	Identify Piping to be Removed	5 days	Fri 12/18/26	Thu 12/24/26									
310	Cut/Drain/Cap Piping to be removed	5 days	Fri 12/25/26	Thu 12/31/26									
311	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 1/1/27	Thu 1/7/27									
312	Make new Connection to Exterior Systems	5 days	Fri 1/8/27	Thu 1/14/27									
313	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 1/15/27	Thu 1/21/27									
314	South	30 days	Fri 12/18/26	Thu 1/28/27									
315	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 12/18/26	Thu 12/24/26									
316	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 12/25/26	Thu 12/31/26									
317	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 1/7/27	Wed 1/13/27									
318	Abandoned Piping: Fill & Cap as Required	5 days	Thu 1/14/27	Wed 1/20/27									
319	Identify Piping to be Removed	5 days	Fri 12/25/26	Thu 12/31/26									
320	Cut/Drain/Cap Piping to be removed	5 days	Fri 1/1/27	Thu 1/7/27									
321	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 1/8/27	Thu 1/14/27									
322	Make new Connection to Exterior Systems	5 days	Fri 1/15/27	Thu 1/21/27									
323	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 1/22/27	Thu 1/28/27									
324	Third Floor	35 days	Fri 12/25/26	Thu 2/11/27									
325	North	30 days	Fri 12/25/26	Thu 2/4/27									
326	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 12/25/26	Thu 12/31/26									
327	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 1/1/27	Thu 1/7/27									
328	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 1/14/27	Wed 1/20/27									
329	Abandoned Piping: Fill & Cap as Required	5 days	Thu 1/21/27	Wed 1/27/27									
330	Identify Piping to be Removed	5 days	Fri 1/1/27	Thu 1/7/27									
331	Cut/Drain/Cap Piping to be removed	5 days	Fri 1/8/27	Thu 1/14/27									
332	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 1/15/27	Thu 1/21/27									
333	Make new Connection to Exterior Systems	5 days	Fri 1/22/27	Thu 1/28/27									
334	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 1/29/27	Thu 2/4/27									
335	South	30 days	Fri 1/1/27	Thu 2/11/27									
336	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 1/1/27	Thu 1/7/27									
337	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 1/8/27	Thu 1/14/27									
338	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 1/21/27	Wed 1/27/27									
339	Abandoned Piping: Fill & Cap as Required	5 days	Thu 1/28/27	Wed 2/3/27									
340	Identify Piping to be Removed	5 days	Fri 1/8/27	Thu 1/14/27									
341	Cut/Drain/Cap Piping to be removed	5 days	Fri 1/15/27	Thu 1/21/27									
342	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 1/22/27	Thu 1/28/27									
343	Make new Connection to Exterior Systems	5 days	Fri 1/29/27	Thu 2/4/27									
344	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 2/5/27	Thu 2/11/27									
345	Attic	35 days	Fri 1/8/27	Thu 2/25/27									
346	North	30 days	Fri 1/8/27	Thu 2/18/27									
347	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 1/8/27	Thu 1/14/27									
348	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 1/15/27	Thu 1/21/27									
349	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 1/28/27	Wed 2/3/27									
350	Abandoned Piping: Fill & Cap as Required	5 days	Thu 2/4/27	Wed 2/10/27									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

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Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
351 Identify Piping to be Removed	5 days	Fri 1/15/27	Thu 1/21/27									
352 Cut/Drain/Cap Piping to be removed	5 days	Fri 1/22/27	Thu 1/28/27									
353 Remove Cut Piping & Replace with Updated Piping	5 days	Fri 1/29/27	Thu 2/4/27									
354 Make new Connection to Exterior Systems	5 days	Fri 2/5/27	Thu 2/11/27									
355 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 2/12/27	Thu 2/18/27									
356 South	30 days	Fri 1/15/27	Thu 2/25/27									
357 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 1/15/27	Thu 1/21/27									
358 Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 1/22/27	Thu 1/28/27									
359 Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 2/4/27	Wed 2/10/27									
360 Abandoned Piping: Fill & Cap as Required	5 days	Thu 2/11/27	Wed 2/17/27									
361 Identify Piping to be Removed	5 days	Fri 1/22/27	Thu 1/28/27									
362 Cut/Drain/Cap Piping to be removed	5 days	Fri 1/29/27	Thu 2/4/27									
363 Remove Cut Piping & Replace with Updated Piping	5 days	Fri 2/5/27	Thu 2/11/27									
364 Make new Connection to Exterior Systems	5 days	Fri 2/12/27	Thu 2/18/27									
365 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 2/19/27	Thu 2/25/27									
366 Phase 01F Main Building - East Range	65 days	Fri 2/26/27	Thu 5/27/27									
367 Setup Work Area and Dust Controls	3 days	Fri 2/26/27	Tue 3/2/27									
368 Tunnel	25 days	Wed 3/3/27	Tue 4/6/27									
369 North	25 days	Wed 3/3/27	Tue 4/6/27									
370 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	2 days	Wed 3/3/27	Thu 3/4/27									
371 Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 3/5/27	Thu 3/11/27									
372 Identify Piping to be "Abandoned: for Future Projects	5 days	Wed 3/3/27	Tue 3/9/27									
373 Abandoned Piping: Fill & Cap as Required	5 days	Wed 3/10/27	Tue 3/16/27									
374 Identify Piping to be Removed	5 days	Fri 3/5/27	Thu 3/11/27									
375 Cut/Drain/Cap Piping to be removed	5 days	Fri 3/12/27	Thu 3/18/27									
376 Remove Cut Piping & Replace with Updated Piping	5 days	Fri 3/19/27	Thu 3/25/27									
377 Make new Connection to Exterior Systems	3 days	Fri 3/26/27	Tue 3/30/27									
378 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 3/31/27	Tue 4/6/27									
379 South	21 days	Fri 3/5/27	Fri 4/2/27									
380 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	2 days	Fri 3/5/27	Mon 3/8/27									
381 Identify existing Pipe to Remain; Do Not Cut	5 days	Tue 3/9/27	Mon 3/15/27									
382 Identify Piping to be "Abandoned: for Future Projects	5 days	Wed 3/10/27	Tue 3/16/27									
383 Abandoned Piping: Fill & Cap as Required	5 days	Wed 3/17/27	Tue 3/23/27									
384 Identify Piping to be Removed	5 days	Tue 3/9/27	Mon 3/15/27									
385 Cut/Drain/Cap Piping to be removed	2 days	Tue 3/16/27	Wed 3/17/27									
386 Remove Cut Piping & Replace with Updated Piping	5 days	Thu 3/18/27	Wed 3/24/27									
387 Make new Connection to Exterior Systems	5 days	Thu 3/25/27	Wed 3/31/27									
388 Patch & Repair Ceiling/Wall/Floor Sections	2 days	Thu 4/1/27	Fri 4/2/27									
389 Ground Floor	31 days	Tue 3/9/27	Tue 4/20/27									
390 North	28 days	Tue 3/9/27	Thu 4/15/27									
391 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 3/9/27	Thu 3/11/27									
392 Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 3/12/27	Thu 3/18/27									
393 Identify Piping to be "Abandoned: for Future Projects	5 days	Wed 3/17/27	Tue 3/23/27									
394 Abandoned Piping: Fill & Cap as Required	5 days	Wed 3/24/27	Tue 3/30/27									
395 Identify Piping to be Removed	5 days	Fri 3/12/27	Thu 3/18/27									
396 Cut/Drain/Cap Piping to be removed	5 days	Fri 3/19/27	Thu 3/25/27									
397 Remove Cut Piping & Replace with Updated Piping	5 days	Fri 3/26/27	Thu 4/1/27									
398 Exterior Utilities	5 days	Fri 4/2/27	Thu 4/8/27									
399 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 4/9/27	Thu 4/15/27									
400 South	28 days	Fri 3/12/27	Tue 4/20/27									

Upgrade Sanitary & Storm Systems 8/19/25	Task Split Milestone	Summary Project Summary Inactive Task	Inactive Milestone Inactive Summary Manual Task	Duration-only Manual Summary Rollup Manual Summary	Start-only Finish-only External Tasks	External Milestone Deadline Progress	Manual Progress
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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
401	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Fri 3/12/27	Tue 3/16/27													
402	Identify existing Pipe to Remain; Do Not Cut	5 days	Wed 3/17/27	Tue 3/23/27													
403	Identify Piping to be "Abandoned: for Future Projects	5 days	Wed 3/24/27	Tue 3/30/27													
404	Abandoned Piping: Fill & Cap as Required	5 days	Wed 3/31/27	Tue 4/6/27													
405	Identify Piping to be Removed	5 days	Wed 3/17/27	Tue 3/23/27													
406	Cut/Drain/Cap Piping to be removed	5 days	Wed 3/24/27	Tue 3/30/27													
407	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 3/31/27	Tue 4/6/27													
408	Exterior Utilities	5 days	Wed 4/7/27	Tue 4/13/27													
409	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 4/14/27	Tue 4/20/27													
410	Mezzanine	28 days	Wed 3/17/27	Fri 4/23/27													
411	North	28 days	Wed 3/17/27	Fri 4/23/27													
412	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 3/17/27	Fri 3/19/27													
413	Identify existing Pipe to Remain; Do Not Cut	5 days	Mon 3/22/27	Fri 3/26/27													
414	Identify Piping to be "Abandoned: for Future Projects	5 days	Wed 3/31/27	Tue 4/6/27													
415	Abandoned Piping: Fill & Cap as Required	5 days	Wed 4/7/27	Tue 4/13/27													
416	Identify Piping to be Removed	5 days	Mon 3/22/27	Fri 3/26/27													
417	Cut/Drain/Cap Piping to be removed	5 days	Mon 3/29/27	Fri 4/2/27													
418	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 4/5/27	Fri 4/9/27													
419	Make new Connection to Exterior Systems	5 days	Mon 4/12/27	Fri 4/16/27													
420	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 4/19/27	Fri 4/23/27													
421	South	22 days	Mon 3/22/27	Tue 4/20/27													
422	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 3/22/27	Wed 3/24/27													
423	Identify existing Pipe to Remain; Do Not Cut	4 days	Thu 3/25/27	Tue 3/30/27													
424	Identify Piping to be "Abandoned: for Future Projects	4 days	Wed 4/7/27	Mon 4/12/27													
425	Abandoned Piping: Fill & Cap as Required	4 days	Tue 4/13/27	Fri 4/16/27													
426	Identify Piping to be Removed	4 days	Thu 3/25/27	Tue 3/30/27													
427	Cut/Drain/Cap Piping to be removed	4 days	Wed 3/31/27	Mon 4/5/27													
428	Remove Cut Piping & Replace with Updated Piping	4 days	Tue 4/6/27	Fri 4/9/27													
429	Make new Connection to Exterior Systems	4 days	Mon 4/12/27	Thu 4/15/27													
430	Patch & Repair Ceiling/Wall/Floor Sections	3 days	Fri 4/16/27	Tue 4/20/27													
431	First Floor	31 days	Thu 3/25/27	Thu 5/6/27													
432	North	28 days	Thu 3/25/27	Mon 5/3/27													
433	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 3/25/27	Mon 3/29/27													
434	Identify existing Pipe to Remain; Do Not Cut	4 days	Tue 3/30/27	Fri 4/2/27													
435	Identify Piping to be "Abandoned: for Future Projects	4 days	Tue 4/13/27	Fri 4/16/27													
436	Abandoned Piping: Fill & Cap as Required	4 days	Mon 4/19/27	Thu 4/22/27													
437	Identify Piping to be Removed	5 days	Tue 3/30/27	Mon 4/5/27													
438	Cut/Drain/Cap Piping to be removed	5 days	Tue 4/6/27	Mon 4/12/27													
439	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 4/13/27	Mon 4/19/27													
440	Make new Connection to Exterior Systems	5 days	Tue 4/20/27	Mon 4/26/27													
441	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 4/27/27	Mon 5/3/27													
442	South	28 days	Tue 3/30/27	Thu 5/6/27													
443	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 3/30/27	Thu 4/1/27													
444	Identify existing Pipe to Remain; Do Not Cut	4 days	Fri 4/2/27	Wed 4/7/27													
445	Identify Piping to be "Abandoned: for Future Projects	4 days	Mon 4/19/27	Thu 4/22/27													
446	Abandoned Piping: Fill & Cap as Required	4 days	Fri 4/23/27	Wed 4/28/27													
447	Identify Piping to be Removed	5 days	Fri 4/2/27	Thu 4/8/27													
448	Cut/Drain/Cap Piping to be removed	5 days	Fri 4/9/27	Thu 4/15/27													
449	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 4/16/27	Thu 4/22/27													
450	Make new Connection to Exterior Systems	5 days	Fri 4/23/27	Thu 4/29/27													
Upgrade Sanitary & Storm Systems 8/19/25					Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress												
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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
451	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 4/30/27	Thu 5/6/27													
452	Second Floor	31 days	Fri 4/2/27	Fri 5/14/27													
453	North	28 days	Fri 4/2/27	Tue 5/11/27													
454	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Fri 4/2/27	Tue 4/6/27													
455	Identify existing Pipe to Remain; Do Not Cut	5 days	Wed 4/7/27	Tue 4/13/27													
456	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 4/23/27	Thu 4/29/27													
457	Abandoned Piping: Fill & Cap as Required	5 days	Fri 4/30/27	Thu 5/6/27													
458	Identify Piping to be Removed	5 days	Wed 4/7/27	Tue 4/13/27													
459	Cut/Drain/Cap Piping to be removed	5 days	Wed 4/14/27	Tue 4/20/27													
460	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 4/21/27	Tue 4/27/27													
461	Make new Connection to Exterior Systems	5 days	Wed 4/28/27	Tue 5/4/27													
462	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 5/5/27	Tue 5/11/27													
463	South	28 days	Wed 4/7/27	Fri 5/14/27													
464	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 4/7/27	Fri 4/9/27													
465	Identify existing Pipe to Remain; Do Not Cut	5 days	Mon 4/12/27	Fri 4/16/27													
466	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 4/30/27	Thu 5/6/27													
467	Abandoned Piping: Fill & Cap as Required	5 days	Fri 5/7/27	Thu 5/13/27													
468	Identify Piping to be Removed	5 days	Mon 4/12/27	Fri 4/16/27													
469	Cut/Drain/Cap Piping to be removed	5 days	Mon 4/19/27	Fri 4/23/27													
470	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 4/26/27	Fri 4/30/27													
471	Make new Connection to Exterior Systems	5 days	Mon 5/3/27	Fri 5/7/27													
472	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 5/10/27	Fri 5/14/27													
473	Third Floor	34 days	Mon 4/12/27	Thu 5/27/27													
474	North	29 days	Mon 4/12/27	Thu 5/20/27													
475	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 4/12/27	Wed 4/14/27													
476	Identify existing Pipe to Remain; Do Not Cut	5 days	Thu 4/15/27	Wed 4/21/27													
477	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 5/7/27	Thu 5/13/27													
478	Abandoned Piping: Fill & Cap as Required	5 days	Fri 5/14/27	Thu 5/20/27													
479	Identify Piping to be Removed	5 days	Thu 4/15/27	Wed 4/21/27													
480	Cut/Drain/Cap Piping to be removed	5 days	Thu 4/22/27	Wed 4/28/27													
481	Remove Cut Piping & Replace with Updated Piping	5 days	Thu 4/29/27	Wed 5/5/27													
482	Make new Connection to Exterior Systems	5 days	Thu 5/6/27	Wed 5/12/27													
483	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 5/13/27	Wed 5/19/27													
484	South	31 days	Thu 4/15/27	Thu 5/27/27													
485	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 4/15/27	Mon 4/19/27													
486	Identify existing Pipe to Remain; Do Not Cut	5 days	Tue 4/20/27	Mon 4/26/27													
487	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 5/14/27	Thu 5/20/27													
488	Abandoned Piping: Fill & Cap as Required	5 days	Fri 5/21/27	Thu 5/27/27													
489	Identify Piping to be Removed	5 days	Tue 4/20/27	Mon 4/26/27													
490	Cut/Drain/Cap Piping to be removed	5 days	Tue 4/27/27	Mon 5/3/27													
491	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 5/4/27	Mon 5/10/27													
492	Make new Connection to Exterior Systems	3 days	Tue 5/11/27	Thu 5/13/27													
493	Patch & Repair Ceiling/Wall/Floor Sections	4 days	Fri 5/14/27	Wed 5/19/27													
494	Phase 01G Main Building - West Range	75 days	Thu 5/20/27	Wed 9/1/27													
495	Setup Work Area and Dust Controls	4 days	Thu 5/20/27	Tue 5/25/27													
496	Tunnel	31 days	Wed 5/26/27	Wed 7/7/27													
497	North	28 days	Wed 5/26/27	Fri 7/2/27													
498	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 5/26/27	Fri 5/28/27													
499	Identify existing Pipe to Remain; Do Not Cut	4 days	Mon 5/31/27	Thu 6/3/27													
500	Identify Piping to be "Abandoned: for Future Projects	4 days	Mon 5/31/27	Thu 6/3/27													
Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress																	
Upgrade Sanitary & Storm Systems 8/19/25																	

Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
501 Abandoned Piping: Fill & Cap as Required	5 days	Fri 6/4/27	Thu 6/10/27									
502 Identify Piping to be Removed	5 days	Mon 5/31/27	Fri 6/4/27									
503 Cut/Drain/Cap Piping to be removed	5 days	Mon 6/7/27	Fri 6/11/27									
504 Remove Cut Piping & Replace with Updated Piping	5 days	Mon 6/14/27	Fri 6/18/27									
505 Make new Connection to Exterior Systems	5 days	Mon 6/21/27	Fri 6/25/27									
506 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 6/28/27	Fri 7/2/27									
507 South	28 days	Mon 5/31/27	Wed 7/7/27									
508 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 5/31/27	Wed 6/2/27									
509 Identify existing Pipe to Remain; Do Not Cut	4 days	Thu 6/3/27	Tue 6/8/27									
510 Identify Piping to be "Abandoned: for Future Projects	4 days	Fri 6/4/27	Wed 6/9/27									
511 Abandoned Piping: Fill & Cap as Required	5 days	Thu 6/10/27	Wed 6/16/27									
512 Identify Piping to be Removed	5 days	Thu 6/3/27	Wed 6/9/27									
513 Cut/Drain/Cap Piping to be removed	5 days	Thu 6/10/27	Wed 6/16/27									
514 Remove Cut Piping & Replace with Updated Piping	5 days	Thu 6/17/27	Wed 6/23/27									
515 Make new Connection to Exterior Systems	5 days	Thu 6/24/27	Wed 6/30/27									
516 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 7/1/27	Wed 7/7/27									
517 Ground Floor	31 days	Thu 6/3/27	Thu 7/15/27									
518 North	28 days	Thu 6/3/27	Mon 7/12/27									
519 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 6/3/27	Mon 6/7/27									
520 Identify existing Pipe to Remain; Do Not Cut	4 days	Tue 6/8/27	Fri 6/11/27									
521 Identify Piping to be "Abandoned: for Future Projects	4 days	Thu 6/10/27	Tue 6/15/27									
522 Abandoned Piping: Fill & Cap as Required	5 days	Wed 6/16/27	Tue 6/22/27									
523 Identify Piping to be Removed	5 days	Tue 6/8/27	Mon 6/14/27									
524 Cut/Drain/Cap Piping to be removed	5 days	Tue 6/15/27	Mon 6/21/27									
525 Remove Cut Piping & Replace with Updated Piping	5 days	Tue 6/22/27	Mon 6/28/27									
526 Exterior Utilities	5 days	Tue 6/29/27	Mon 7/5/27									
527 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 7/6/27	Mon 7/12/27									
528 South	28 days	Tue 6/8/27	Thu 7/15/27									
529 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 6/8/27	Thu 6/10/27									
530 Identify existing Pipe to Remain; Do Not Cut	4 days	Fri 6/11/27	Wed 6/16/27									
531 Identify Piping to be "Abandoned: for Future Projects	4 days	Wed 6/16/27	Mon 6/21/27									
532 Abandoned Piping: Fill & Cap as Required	5 days	Tue 6/22/27	Mon 6/28/27									
533 Identify Piping to be Removed	5 days	Fri 6/11/27	Thu 6/17/27									
534 Cut/Drain/Cap Piping to be removed	5 days	Fri 6/18/27	Thu 6/24/27									
535 Remove Cut Piping & Replace with Updated Piping	5 days	Fri 6/25/27	Thu 7/1/27									
536 Exterior Utilities	5 days	Fri 7/2/27	Thu 7/8/27									
537 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 7/9/27	Thu 7/15/27									
538 Mezzanine	35 days	Fri 6/11/27	Thu 7/29/27									
539 North	30 days	Fri 6/11/27	Thu 7/22/27									
540 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 6/11/27	Thu 6/17/27									
541 Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 6/18/27	Thu 6/24/27									
542 Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 6/22/27	Mon 6/28/27									
543 Abandoned Piping: Fill & Cap as Required	5 days	Tue 6/29/27	Mon 7/5/27									
544 Identify Piping to be Removed	5 days	Fri 6/18/27	Thu 6/24/27									
545 Cut/Drain/Cap Piping to be removed	5 days	Fri 6/25/27	Thu 7/1/27									
546 Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/2/27	Thu 7/8/27									
547 Make new Connection to Exterior Systems	5 days	Fri 7/9/27	Thu 7/15/27									
548 Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 7/16/27	Thu 7/22/27									
549 South	30 days	Fri 6/18/27	Thu 7/29/27									
550 Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 6/18/27	Thu 6/24/27									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
551	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 6/25/27	Thu 7/1/27									
552	Identify Piping to be "Abandoned: for Future Projects	5 days	Tue 6/29/27	Mon 7/5/27									
553	Abandoned Piping: Fill & Cap as Required	5 days	Tue 7/6/27	Mon 7/12/27									
554	Identify Piping to be Removed	5 days	Fri 6/25/27	Thu 7/1/27									
555	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/2/27	Thu 7/8/27									
556	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/9/27	Thu 7/15/27									
557	Make new Connection to Exterior Systems	5 days	Fri 7/16/27	Thu 7/22/27									
558	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 7/23/27	Thu 7/29/27									
559	First Floor	31 days	Fri 6/25/27	Fri 8/6/27									
560	North	28 days	Fri 6/25/27	Tue 8/3/27									
561	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Fri 6/25/27	Tue 6/29/27									
562	Identify existing Pipe to Remain; Do Not Cut	4 days	Wed 6/30/27	Mon 7/5/27									
563	Identify Piping to be "Abandoned: for Future Projects	4 days	Tue 7/6/27	Fri 7/9/27									
564	Abandoned Piping: Fill & Cap as Required	5 days	Mon 7/12/27	Fri 7/16/27									
565	Identify Piping to be Removed	5 days	Wed 6/30/27	Tue 7/6/27									
566	Cut/Drain/Cap Piping to be removed	5 days	Wed 7/7/27	Tue 7/13/27									
567	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 7/14/27	Tue 7/20/27									
568	Make new Connection to Exterior Systems	5 days	Wed 7/21/27	Tue 7/27/27									
569	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 7/28/27	Tue 8/3/27									
570	South	28 days	Wed 6/30/27	Fri 8/6/27									
571	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 6/30/27	Fri 7/2/27									
572	Identify existing Pipe to Remain; Do Not Cut	4 days	Mon 7/5/27	Thu 7/8/27									
573	Identify Piping to be "Abandoned: for Future Projects	4 days	Mon 7/12/27	Thu 7/15/27									
574	Abandoned Piping: Fill & Cap as Required	5 days	Fri 7/16/27	Thu 7/22/27									
575	Identify Piping to be Removed	5 days	Mon 7/5/27	Fri 7/9/27									
576	Cut/Drain/Cap Piping to be removed	5 days	Mon 7/12/27	Fri 7/16/27									
577	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 7/19/27	Fri 7/23/27									
578	Make new Connection to Exterior Systems	5 days	Mon 7/26/27	Fri 7/30/27									
579	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 8/2/27	Fri 8/6/27									
580	Second Floor	31 days	Mon 7/5/27	Mon 8/16/27									
581	North	28 days	Mon 7/5/27	Wed 8/11/27									
582	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 7/5/27	Wed 7/7/27									
583	Identify existing Pipe to Remain; Do Not Cut	4 days	Thu 7/8/27	Tue 7/13/27									
584	Identify Piping to be "Abandoned: for Future Projects	4 days	Fri 7/16/27	Wed 7/21/27									
585	Abandoned Piping: Fill & Cap as Required	5 days	Thu 7/22/27	Wed 7/28/27									
586	Identify Piping to be Removed	5 days	Thu 7/8/27	Wed 7/14/27									
587	Cut/Drain/Cap Piping to be removed	5 days	Thu 7/15/27	Wed 7/21/27									
588	Remove Cut Piping & Replace with Updated Piping	5 days	Thu 7/22/27	Wed 7/28/27									
589	Make new Connection to Exterior Systems	5 days	Thu 7/29/27	Wed 8/4/27									
590	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 8/5/27	Wed 8/11/27									
591	South	28 days	Thu 7/8/27	Mon 8/16/27									
592	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Thu 7/8/27	Mon 7/12/27									
593	Identify existing Pipe to Remain; Do Not Cut	4 days	Tue 7/13/27	Fri 7/16/27									
594	Identify Piping to be "Abandoned: for Future Projects	4 days	Thu 7/22/27	Tue 7/27/27									
595	Abandoned Piping: Fill & Cap as Required	5 days	Wed 7/28/27	Tue 8/3/27									
596	Identify Piping to be Removed	5 days	Tue 7/13/27	Mon 7/19/27									
597	Cut/Drain/Cap Piping to be removed	5 days	Tue 7/20/27	Mon 7/26/27									
598	Remove Cut Piping & Replace with Updated Piping	5 days	Tue 7/27/27	Mon 8/2/27									
599	Make new Connection to Exterior Systems	5 days	Tue 8/3/27	Mon 8/9/27									
600	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Tue 8/10/27	Mon 8/16/27									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter										
601	Third Floor	31 days	Tue 7/13/27	Tue 8/24/27																			
602	North	28 days	Tue 7/13/27	Thu 8/19/27																			
603	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 7/13/27	Thu 7/15/27																			
604	Identify existing Pipe to Remain; Do Not Cut	4 days	Fri 7/16/27	Wed 7/21/27																			
605	Identify Piping to be "Abandoned: for Future Projects	4 days	Wed 7/28/27	Mon 8/2/27																			
606	Abandoned Piping: Fill & Cap as Required	5 days	Tue 8/3/27	Mon 8/9/27																			
607	Identify Piping to be Removed	5 days	Fri 7/16/27	Thu 7/22/27																			
608	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/23/27	Thu 7/29/27																			
609	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/30/27	Thu 8/5/27																			
610	Make new Connection to Exterior Systems	5 days	Fri 8/6/27	Thu 8/12/27																			
611	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 8/13/27	Thu 8/19/27																			
612	South	28 days	Fri 7/16/27	Tue 8/24/27																			
613	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Fri 7/16/27	Tue 7/20/27																			
614	Identify existing Pipe to Remain; Do Not Cut	4 days	Wed 7/21/27	Mon 7/26/27																			
615	Identify Piping to be "Abandoned: for Future Projects	4 days	Tue 8/3/27	Fri 8/6/27																			
616	Abandoned Piping: Fill & Cap as Required	5 days	Mon 8/9/27	Fri 8/13/27																			
617	Identify Piping to be Removed	5 days	Wed 7/21/27	Tue 7/27/27																			
618	Cut/Drain/Cap Piping to be removed	5 days	Wed 7/28/27	Tue 8/3/27																			
619	Remove Cut Piping & Replace with Updated Piping	5 days	Wed 8/4/27	Tue 8/10/27																			
620	Make new Connection to Exterior Systems	5 days	Wed 8/11/27	Tue 8/17/27																			
621	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Wed 8/18/27	Tue 8/24/27																			
622	Attic	31 days	Wed 7/21/27	Wed 9/1/27																			
623	North	28 days	Wed 7/21/27	Fri 8/27/27																			
624	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Wed 7/21/27	Fri 7/23/27																			
625	Identify existing Pipe to Remain; Do Not Cut	4 days	Mon 7/26/27	Thu 7/29/27																			
626	Identify Piping to be "Abandoned: for Future Projects	4 days	Mon 8/9/27	Thu 8/12/27																			
627	Abandoned Piping: Fill & Cap as Required	5 days	Fri 8/13/27	Thu 8/19/27																			
628	Identify Piping to be Removed	5 days	Mon 7/26/27	Fri 7/30/27																			
629	Cut/Drain/Cap Piping to be removed	5 days	Mon 8/2/27	Fri 8/6/27																			
630	Remove Cut Piping & Replace with Updated Piping	5 days	Mon 8/9/27	Fri 8/13/27																			
631	Make new Connection to Exterior Systems	5 days	Mon 8/16/27	Fri 8/20/27																			
632	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Mon 8/23/27	Fri 8/27/27																			
633	South	28 days	Mon 7/26/27	Wed 9/1/27																			
634	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Mon 7/26/27	Wed 7/28/27																			
635	Identify existing Pipe to Remain; Do Not Cut	4 days	Thu 7/29/27	Tue 8/3/27																			
636	Identify Piping to be "Abandoned: for Future Projects	4 days	Fri 8/13/27	Wed 8/18/27																			
637	Abandoned Piping: Fill & Cap as Required	5 days	Thu 8/19/27	Wed 8/25/27																			
638	Identify Piping to be Removed	5 days	Thu 7/29/27	Wed 8/4/27																			
639	Cut/Drain/Cap Piping to be removed	5 days	Thu 8/5/27	Wed 8/11/27																			
640	Remove Cut Piping & Replace with Updated Piping	5 days	Thu 8/12/27	Wed 8/18/27																			
641	Make new Connection to Exterior Systems	5 days	Thu 8/19/27	Wed 8/25/27																			
642	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Thu 8/26/27	Wed 9/1/27																			
643	Phase 2 West Wing	125 days	Thu 9/2/27	Wed 2/23/28																			
644	Setup Work Area and Dust Controls	3 days	Thu 9/2/27	Mon 9/6/27																			
645	Basement	33 days	Tue 9/7/27	Thu 10/21/27																			
646	North	28 days	Tue 9/7/27	Thu 10/14/27																			
647	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	3 days	Tue 9/7/27	Thu 9/9/27																			
648	Identify existing Pipe to Remain; Do Not Cut	2 days	Fri 9/10/27	Mon 9/13/27																			
649	Identify Piping to be "Abandoned: for Future Projects	2 days	Tue 9/7/27	Wed 9/8/27																			
650	Abandoned Piping: Fill & Cap as Required	5 days	Thu 9/9/27	Wed 9/15/27																			
Upgrade Sanitary & Storm Systems 8/19/25														Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress									
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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
651	Identify Piping to be Removed	5 days	Fri 9/10/27	Thu 9/16/27													
652	Cut/Drain/Cap Piping to be removed	5 days	Fri 9/17/27	Thu 9/23/27													
653	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 9/24/27	Thu 9/30/27													
654	Make new Connection to Exterior Systems	5 days	Fri 10/1/27	Thu 10/7/27													
655	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 10/8/27	Thu 10/14/27													
656	South	31 days	Thu 9/9/27	Thu 10/21/27													
657	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 9/10/27	Thu 9/16/27													
658	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 9/17/27	Thu 9/23/27													
659	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 9/9/27	Wed 9/15/27													
660	Abandoned Piping: Fill & Cap as Required	5 days	Thu 9/16/27	Wed 9/22/27													
661	Identify Piping to be Removed	5 days	Fri 9/17/27	Thu 9/23/27													
662	Cut/Drain/Cap Piping to be removed	5 days	Fri 9/24/27	Thu 9/30/27													
663	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 10/1/27	Thu 10/7/27													
664	Make new Connection to Exterior Systems	5 days	Fri 10/8/27	Thu 10/14/27													
665	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 10/15/27	Thu 10/21/27													
666	Ground Floor	36 days	Thu 9/16/27	Thu 11/4/27													
667	North	31 days	Thu 9/16/27	Thu 10/28/27													
668	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 9/17/27	Thu 9/23/27													
669	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 9/24/27	Thu 9/30/27													
670	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 9/16/27	Wed 9/22/27													
671	Abandoned Piping: Fill & Cap as Required	5 days	Thu 9/23/27	Wed 9/29/27													
672	Identify Piping to be Removed	5 days	Fri 9/24/27	Thu 9/30/27													
673	Cut/Drain/Cap Piping to be removed	5 days	Fri 10/1/27	Thu 10/7/27													
674	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 10/8/27	Thu 10/14/27													
675	Exterior Utilities	5 days	Fri 10/15/27	Thu 10/21/27													
676	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 10/22/27	Thu 10/28/27													
677	South	31 days	Thu 9/23/27	Thu 11/4/27													
678	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 9/24/27	Thu 9/30/27													
679	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 10/1/27	Thu 10/7/27													
680	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 9/23/27	Wed 9/29/27													
681	Abandoned Piping: Fill & Cap as Required	5 days	Thu 9/30/27	Wed 10/6/27													
682	Identify Piping to be Removed	5 days	Fri 10/1/27	Thu 10/7/27													
683	Cut/Drain/Cap Piping to be removed	5 days	Fri 10/8/27	Thu 10/14/27													
684	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 10/15/27	Thu 10/21/27													
685	Exterior Utilities	5 days	Fri 10/22/27	Thu 10/28/27													
686	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 10/29/27	Thu 11/4/27													
687	First Floor	36 days	Thu 9/30/27	Thu 11/18/27													
688	North	31 days	Thu 9/30/27	Thu 11/11/27													
689	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 10/1/27	Thu 10/7/27													
690	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 10/8/27	Thu 10/14/27													
691	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 9/30/27	Wed 10/6/27													
692	Abandoned Piping: Fill & Cap as Required	5 days	Thu 10/7/27	Wed 10/13/27													
693	Identify Piping to be Removed	5 days	Fri 10/8/27	Thu 10/14/27													
694	Cut/Drain/Cap Piping to be removed	5 days	Fri 10/15/27	Thu 10/21/27													
695	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 10/22/27	Thu 10/28/27													
696	Make new Connection to Exterior Systems	5 days	Fri 10/29/27	Thu 11/4/27													
697	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 11/5/27	Thu 11/11/27													
698	South	31 days	Thu 10/7/27	Thu 11/18/27													
699	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 10/8/27	Thu 10/14/27													
700	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 10/15/27	Thu 10/21/27													
Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress																	
Upgrade Sanitary & Storm Systems 8/19/25																	

ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
701	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 10/7/27	Wed 10/13/27									
702	Abandoned Piping: Fill & Cap as Required	5 days	Thu 10/14/27	Wed 10/20/27									
703	Identify Piping to be Removed	5 days	Fri 10/15/27	Thu 10/21/27									
704	Cut/Drain/Cap Piping to be removed	5 days	Fri 10/22/27	Thu 10/28/27									
705	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 10/29/27	Thu 11/4/27									
706	Make new Connection to Exterior Systems	5 days	Fri 11/5/27	Thu 11/11/27									
707	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 11/12/27	Thu 11/18/27									
708	Second Floor	36 days	Thu 10/14/27	Thu 12/2/27									
709	North	31 days	Thu 10/14/27	Thu 11/25/27									
710	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 10/15/27	Thu 10/21/27									
711	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 10/22/27	Thu 10/28/27									
712	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 10/14/27	Wed 10/20/27									
713	Abandoned Piping: Fill & Cap as Required	5 days	Thu 10/21/27	Wed 10/27/27									
714	Identify Piping to be Removed	5 days	Fri 10/22/27	Thu 10/28/27									
715	Cut/Drain/Cap Piping to be removed	5 days	Fri 10/29/27	Thu 11/4/27									
716	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 11/5/27	Thu 11/11/27									
717	Make new Connection to Exterior Systems	5 days	Fri 11/12/27	Thu 11/18/27									
718	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 11/19/27	Thu 11/25/27									
719	South	31 days	Thu 10/21/27	Thu 12/2/27									
720	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 10/22/27	Thu 10/28/27									
721	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 10/29/27	Thu 11/4/27									
722	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 10/21/27	Wed 10/27/27									
723	Abandoned Piping: Fill & Cap as Required	5 days	Thu 10/28/27	Wed 11/3/27									
724	Identify Piping to be Removed	5 days	Fri 10/29/27	Thu 11/4/27									
725	Cut/Drain/Cap Piping to be removed	5 days	Fri 11/5/27	Thu 11/11/27									
726	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 11/12/27	Thu 11/18/27									
727	Make new Connection to Exterior Systems	5 days	Fri 11/19/27	Thu 11/25/27									
728	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 11/26/27	Thu 12/2/27									
729	Third Floor	36 days	Thu 10/28/27	Thu 12/16/27									
730	North	31 days	Thu 10/28/27	Thu 12/9/27									
731	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 10/29/27	Thu 11/4/27									
732	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 11/5/27	Thu 11/11/27									
733	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 10/28/27	Wed 11/3/27									
734	Abandoned Piping: Fill & Cap as Required	5 days	Thu 11/4/27	Wed 11/10/27									
735	Identify Piping to be Removed	5 days	Fri 11/5/27	Thu 11/11/27									
736	Cut/Drain/Cap Piping to be removed	5 days	Fri 11/12/27	Thu 11/18/27									
737	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 11/19/27	Thu 11/25/27									
738	Make new Connection to Exterior Systems	5 days	Fri 11/26/27	Thu 12/2/27									
739	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 12/3/27	Thu 12/9/27									
740	South	31 days	Thu 11/4/27	Thu 12/16/27									
741	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 11/5/27	Thu 11/11/27									
742	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 11/12/27	Thu 11/18/27									
743	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 11/4/27	Wed 11/10/27									
744	Abandoned Piping: Fill & Cap as Required	5 days	Thu 11/11/27	Wed 11/17/27									
745	Identify Piping to be Removed	5 days	Fri 11/12/27	Thu 11/18/27									
746	Cut/Drain/Cap Piping to be removed	5 days	Fri 11/19/27	Thu 11/25/27									
747	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 11/26/27	Thu 12/2/27									
748	Make new Connection to Exterior Systems	5 days	Fri 12/3/27	Thu 12/9/27									
749	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 12/10/27	Thu 12/16/27									
750	Fourth Floor	36 days	Thu 11/11/27	Thu 12/30/27									

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

External Milestone

Deadline

Progress

Manual Progress

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Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
751 North	31 days	Thu 11/11/27	Thu 12/23/27													
752	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 11/12/27	Thu 11/18/27												
753	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 11/19/27	Thu 11/25/27												
754	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 11/11/27	Wed 11/17/27												
755	Abandoned Piping: Fill & Cap as Required	5 days	Thu 11/18/27	Wed 11/24/27												
756	Identify Piping to be Removed	5 days	Fri 11/19/27	Thu 11/25/27												
757	Cut/Drain/Cap Piping to be removed	5 days	Fri 11/26/27	Thu 12/2/27												
758	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 12/3/27	Thu 12/9/27												
759	Make new Connection to Exterior Systems	5 days	Fri 12/10/27	Thu 12/16/27												
760	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 12/17/27	Thu 12/23/27												
761	South	31 days	Thu 11/18/27	Thu 12/30/27												
762	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 11/19/27	Thu 11/25/27												
763	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 11/26/27	Thu 12/2/27												
764	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 11/18/27	Wed 11/24/27												
765	Abandoned Piping: Fill & Cap as Required	5 days	Thu 11/25/27	Wed 12/1/27												
766	Identify Piping to be Removed	5 days	Fri 11/26/27	Thu 12/2/27												
767	Cut/Drain/Cap Piping to be removed	5 days	Fri 12/3/27	Thu 12/9/27												
768	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 12/10/27	Thu 12/16/27												
769	Make new Connection to Exterior Systems	5 days	Fri 12/17/27	Thu 12/23/27												
770	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 12/24/27	Thu 12/30/27												
771	Fifth Floor	65 days	Thu 11/25/27	Wed 2/23/28												
772	North	31 days	Thu 11/25/27	Thu 1/6/28												
773	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 11/26/27	Thu 12/2/27												
774	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 12/3/27	Thu 12/9/27												
775	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 11/25/27	Wed 12/1/27												
776	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/2/27	Wed 12/8/27												
777	Identify Piping to be Removed	5 days	Fri 12/3/27	Thu 12/9/27												
778	Cut/Drain/Cap Piping to be removed	5 days	Fri 12/10/27	Thu 12/16/27												
779	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 12/17/27	Thu 12/23/27												
780	Make new Connection to Exterior Systems	5 days	Fri 12/24/27	Thu 12/30/27												
781	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 12/31/27	Thu 1/6/28												
782	South	60 days	Thu 12/2/27	Wed 2/23/28												
783	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 12/3/27	Thu 12/9/27												
784	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 12/10/27	Thu 12/16/27												
785	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/2/27	Wed 12/8/27												
786	Abandoned Piping: Fill & Cap as Required	55 days	Thu 12/9/27	Wed 2/23/28												
787	Identify Piping to be Removed	5 days	Fri 12/10/27	Thu 12/16/27												
788	Cut/Drain/Cap Piping to be removed	5 days	Fri 12/17/27	Thu 12/23/27												
789	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 12/24/27	Thu 12/30/27												
790	Make new Connection to Exterior Systems	5 days	Fri 12/31/27	Thu 1/6/28												
791	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 1/7/28	Thu 1/13/28												
792	Sixth Floor	36 days	Thu 12/9/27	Thu 1/27/28												
793	North	31 days	Thu 12/9/27	Thu 1/20/28												
794	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 12/10/27	Thu 12/16/27												
795	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 12/17/27	Thu 12/23/27												
796	Identify Piping to be "Abandoned: for Future Projects	5 days	Thu 12/9/27	Wed 12/15/27												
797	Abandoned Piping: Fill & Cap as Required	5 days	Thu 12/16/27	Wed 12/22/27												
798	Identify Piping to be Removed	5 days	Fri 12/17/27	Thu 12/23/27												
799	Cut/Drain/Cap Piping to be removed	5 days	Fri 12/24/27	Thu 12/30/27												
800	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 12/31/27	Thu 1/6/28												

Upgrade Sanitary & Storm Systems
8/19/25

Task

Split

Milestone

Summary

Project Summary

Inactive Task

Inactive Milestone

Inactive Summary

Manual Task

Duration-only

Manual Summary Rollup

Manual Summary

Start-only

Finish-only

External Tasks

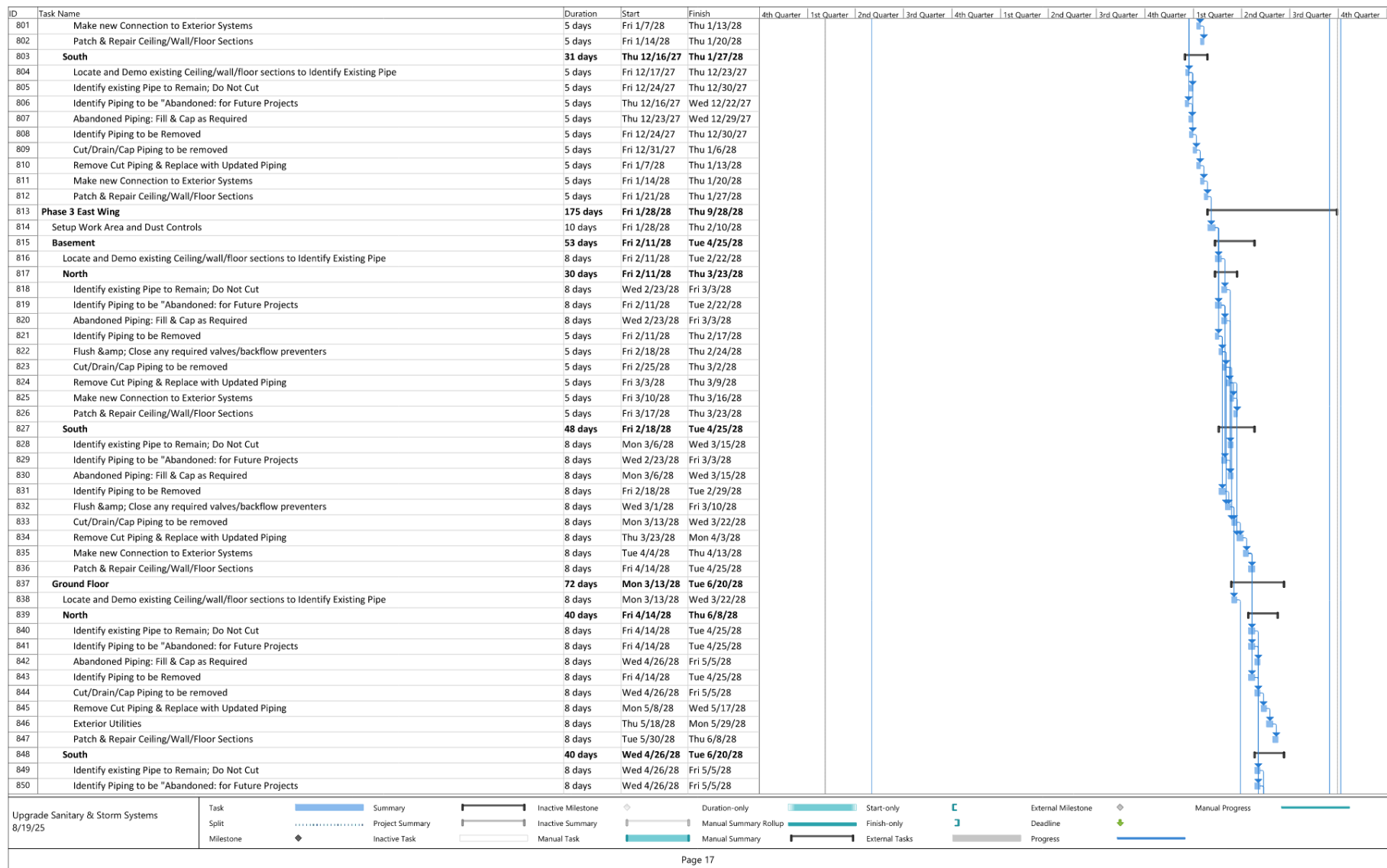
External Milestone

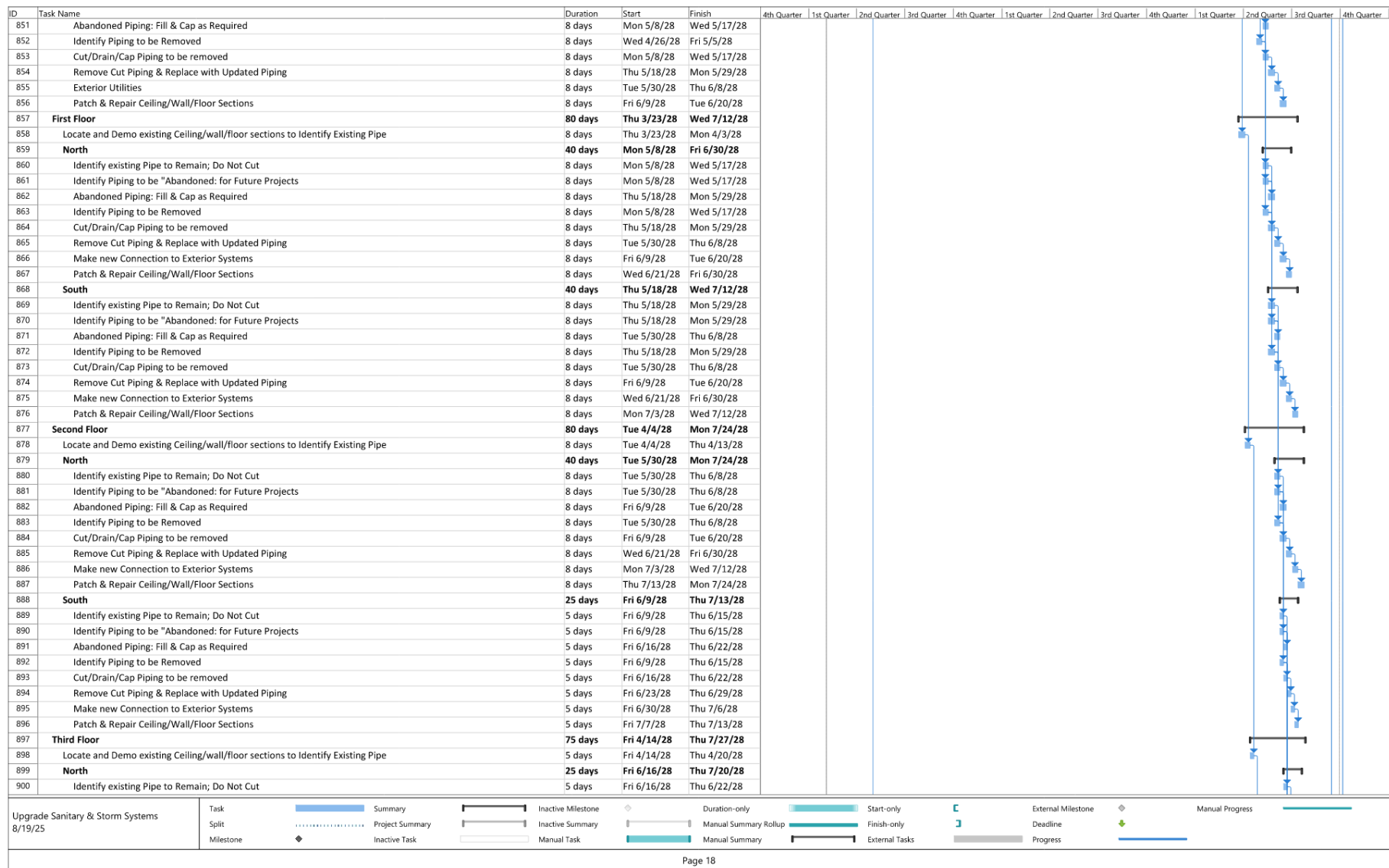
Deadline

Progress

Manual Progress

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ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
901	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 6/16/28	Thu 6/22/28													
902	Abandoned Piping: Fill & Cap as Required	5 days	Fri 6/23/28	Thu 6/29/28													
903	Identify Piping to be Removed	5 days	Fri 6/16/28	Thu 6/22/28													
904	Cut/Drain/Cap Piping to be removed	5 days	Fri 6/23/28	Thu 6/29/28													
905	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 6/30/28	Thu 7/6/28													
906	Make new Connection to Exterior Systems	5 days	Fri 7/7/28	Thu 7/13/28													
907	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 7/14/28	Thu 7/20/28													
908	South	25 days	Fri 6/23/28	Thu 7/27/28													
909	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 6/23/28	Thu 6/29/28													
910	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 6/23/28	Thu 6/29/28													
911	Abandoned Piping: Fill & Cap as Required	5 days	Fri 6/30/28	Thu 7/6/28													
912	Identify Piping to be Removed	5 days	Fri 6/23/28	Thu 6/29/28													
913	Cut/Drain/Cap Piping to be removed	5 days	Fri 6/30/28	Thu 7/6/28													
914	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/7/28	Thu 7/13/28													
915	Make new Connection to Exterior Systems	5 days	Fri 7/14/28	Thu 7/20/28													
916	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 7/21/28	Thu 7/27/28													
917	Fourth Floor	80 days	Fri 4/21/28	Thu 8/10/28													
918	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 4/21/28	Thu 4/27/28													
919	North	25 days	Fri 6/30/28	Thu 8/3/28													
920	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 6/30/28	Thu 7/6/28													
921	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 6/30/28	Thu 7/6/28													
922	Abandoned Piping: Fill & Cap as Required	5 days	Fri 7/7/28	Thu 7/13/28													
923	Identify Piping to be Removed	5 days	Fri 6/30/28	Thu 7/6/28													
924	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/7/28	Thu 7/13/28													
925	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/14/28	Thu 7/20/28													
926	Make new Connection to Exterior Systems	5 days	Fri 7/21/28	Thu 7/27/28													
927	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 7/28/28	Thu 8/3/28													
928	South	25 days	Fri 7/7/28	Thu 8/10/28													
929	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 7/7/28	Thu 7/13/28													
930	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 7/7/28	Thu 7/13/28													
931	Abandoned Piping: Fill & Cap as Required	5 days	Fri 7/14/28	Thu 7/20/28													
932	Identify Piping to be Removed	5 days	Fri 7/7/28	Thu 7/13/28													
933	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/14/28	Thu 7/20/28													
934	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/21/28	Thu 7/27/28													
935	Make new Connection to Exterior Systems	5 days	Fri 7/28/28	Thu 8/3/28													
936	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 8/4/28	Thu 8/10/28													
937	Fifth Floor	85 days	Fri 4/28/28	Thu 8/24/28													
938	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 4/28/28	Thu 5/4/28													
939	North	25 days	Fri 7/14/28	Thu 8/17/28													
940	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 7/14/28	Thu 7/20/28													
941	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 7/14/28	Thu 7/20/28													
942	Abandoned Piping: Fill & Cap as Required	5 days	Fri 7/21/28	Thu 7/27/28													
943	Identify Piping to be Removed	5 days	Fri 7/14/28	Thu 7/20/28													
944	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/21/28	Thu 7/27/28													
945	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 7/28/28	Thu 8/3/28													
946	Make new Connection to Exterior Systems	5 days	Fri 8/4/28	Thu 8/10/28													
947	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 8/11/28	Thu 8/17/28													
948	South	25 days	Fri 7/21/28	Thu 8/24/28													
949	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 7/21/28	Thu 7/27/28													
950	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 7/21/28	Thu 7/27/28													
Upgrade Sanitary & Storm Systems 8/19/25 Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress																	

ID	Task Name	Duration	Start	Finish	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
951	Abandoned Piping: Fill & Cap as Required	5 days	Fri 7/28/28	Thu 8/3/28													
952	Identify Piping to be Removed	5 days	Fri 7/21/28	Thu 7/27/28													
953	Cut/Drain/Cap Piping to be removed	5 days	Fri 7/28/28	Thu 8/3/28													
954	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 8/4/28	Thu 8/10/28													
955	Make new Connection to Exterior Systems	5 days	Fri 8/11/28	Thu 8/17/28													
956	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 8/18/28	Thu 8/24/28													
957	Sixth Floor	90 days	Fri 5/5/28	Thu 9/7/28													
958	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 5/5/28	Thu 5/11/28													
959	North	25 days	Fri 7/28/28	Thu 8/31/28													
960	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 7/28/28	Thu 8/3/28													
961	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 7/28/28	Thu 8/3/28													
962	Abandoned Piping: Fill & Cap as Required	5 days	Fri 8/4/28	Thu 8/10/28													
963	Identify Piping to be Removed	5 days	Fri 7/28/28	Thu 8/3/28													
964	Cut/Drain/Cap Piping to be removed	5 days	Fri 8/4/28	Thu 8/10/28													
965	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 8/11/28	Thu 8/17/28													
966	Make new Connection to Exterior Systems	5 days	Fri 8/18/28	Thu 8/24/28													
967	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 8/25/28	Thu 8/31/28													
968	South	25 days	Fri 8/4/28	Thu 9/7/28													
969	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 8/4/28	Thu 8/10/28													
970	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 8/4/28	Thu 8/10/28													
971	Abandoned Piping: Fill & Cap as Required	5 days	Fri 8/11/28	Thu 8/17/28													
972	Identify Piping to be Removed	5 days	Fri 8/4/28	Thu 8/10/28													
973	Cut/Drain/Cap Piping to be removed	5 days	Fri 8/11/28	Thu 8/17/28													
974	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 8/18/28	Thu 8/24/28													
975	Make new Connection to Exterior Systems	5 days	Fri 8/25/28	Thu 8/31/28													
976	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 9/1/28	Thu 9/7/28													
977	Penthouse	25 days	Fri 8/11/28	Thu 9/14/28													
978	Identify existing Pipe to Remain; Do Not Cut	5 days	Fri 8/11/28	Thu 8/17/28													
979	Identify Piping to be "Abandoned: for Future Projects	5 days	Fri 8/11/28	Thu 8/17/28													
980	Abandoned Piping: Fill & Cap as Required	5 days	Fri 8/18/28	Thu 8/24/28													
981	Identify Piping to be Removed	5 days	Fri 8/11/28	Thu 8/17/28													
982	Cut/Drain/Cap Piping to be removed	5 days	Fri 8/18/28	Thu 8/24/28													
983	Remove Cut Piping & Replace with Updated Piping	5 days	Fri 8/25/28	Thu 8/31/28													
984	Make new Connection to Exterior Systems	5 days	Fri 9/1/28	Thu 9/7/28													
985	Patch & Repair Ceiling/Wall/Floor Sections	5 days	Fri 9/1/28	Thu 9/7/28													
986	Locate and Demo existing Ceiling/wall/floor sections to Identify Existing Pipe	5 days	Fri 9/8/28	Thu 9/14/28													
987	Roof	5 days	Fri 9/8/28	Thu 9/14/28													
988	Make Connections to Current Roof System	5 days	Fri 9/8/28	Thu 9/14/28													
989	Demobilize	10 days	Fri 9/15/28	Thu 9/28/28													
990	Option 3	40 days	Mon 4/27/26	Fri 6/19/26													
991	Roof Scupper replacement	40 days	Mon 4/27/26	Fri 6/19/26													
Upgrade Sanitary & Storm Systems 8/19/25 Task Split Milestone Summary Project Summary Inactive Task Inactive Milestone Inactive Summary Manual Task Duration-only Manual Summary Rollup Manual Summary Start-only Finish-only External Tasks External Milestone Deadline Progress Manual Progress																	
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4.0 COST MODEL

4.1 COST ESTIMATE

A cost estimate is being issued under a separate cover. The estimate was developed in CostOS in accordance with SI standards and templates.

5.0 ARCHITECTURE

This Architectural narrative has been updated to reflect the development of the NMNH Upgrade Sanitary and Storm design since the Implementation Study Phase. Please refer to the 100% Construction Documents drawings for additional information.

5.1 REGULATORY CRITERIA

- Smithsonian Institution Facilities Design Standards - 2021 Edition
- International Building Code (IBC) 2021 Edition
- International Existing Building Code (IEBC) 2021 Edition
- (ANSI) American National Standards Institute All related documents.
- Smithsonian Institution, Office of Facilities and Operations, Codes, Standards and Guidelines Latest edition
- Smithsonian Sustainability Plans Latest edition.

5.2 MAIN BUILDING

The ground-breaking ceremony for the new United States National Museum was held on June 15, 1904, and the museum officially opened to the public on March 17, 1910, just before it was fully complete. Over time the building became known as the National Museum of Natural History (NMNH). Over the course of more than a century, the building and site have undergone continuous, substantial changes, resulting in a building that continues to evolve to this day.

5.2.1 TUNNELS

The Tunnels are located below the Ground Floor of the Main Building. They were originally built to accommodate the distribution of Mechanical, Electrical, and Plumbing systems across the building. Over the years the tunnels have become congested in various locations. To accommodate the extensive piping replacement required in the tunnels, a few of the tunnel access hatch locations on the Ground Floor will need to be replaced and enlarged to facilitate work within the tunnels. It is also anticipated that access to the tunnels will require extensive use of the space where the hatch access points are located. This may interfere with the function of the space while the tunnel work is being performed.

5.2.2 NORTH LOBBY

The North Lobby is the Ground Floor access point for staff and the public alike to enter the Museum. Located off the North Lobby, the public restrooms are extensively used by the public.

The interior elements of the North Lobby Restrooms, finishes, fixtures, and toilet accessories have been refreshed over the years due to constant wear and tear. Universally accessible toilet stalls and fixtures have been installed in both North Lobby restrooms. Existing tiling in the North Lobby restrooms produce cleaning and maintenance issues due to their small format and excessive grout lines. The existing tile has been coated to protect the grout and provide surfaces that are easier to clean. The coating strategy has been implemented in several restrooms throughout the building. While this has eased maintenance and provided an aesthetic refresh, recoating does not provide long-term durability.

5.0 ARCHITECTURE

The design intent for the North Lobby Restrooms is to update the current layout and upgrade the fixtures and finishes to provide a higher level of durability. A new Family Restroom, located near the Evans Gallery, includes accessible fixtures, accessories, and an adult changing table. Within North Lobby Restrooms, the tiling, paint, and toilet partitions will be improved to remedy the current hygiene concerns. The sanitary scope for this area will be minimal due to its frequent maintenance over its history, which provides the opportunity to phase the work later.

This will allow the public to continue to use these restrooms while work is happening in non-public areas.

5.2.3 WEST DOCK

The West Dock is a major access point for deliveries to the building. There are two restrooms directly beside the loading dock that will require sanitary piping replacement and updated layouts. Due to its proximity to a vital loading area, the existing perimeter walls will be maintained to minimize disruption. The revised layouts provide universal accessibility while also converting them into single-use restrooms.

5.2.4 THIRD FLOOR WOMEN'S RESTROOM

The Third Floor Women's Restroom is one of the few remaining restrooms from the museum's original early 1900s construction. Finishes in this space include marble flooring and wall coverings, laminate counter tops (Added later), and wooden toilet stalls. The design team will document the salvage and storage of the historic material for NMNH use in the future. The space will receive all new finishes to be aligned with new restroom standards being implemented as part of this project.

5.2.5 THIRD FLOOR PANTRIES

The Third Floor scope includes two new pantries to respond to the removal of office sinks in offices in this portion of the building. One of these pantries will be in the Rose Room. The other pantry will be located outside of Room 317. The design of both pantries provides six-foot layouts, including cabinetry, countertop, sink, a commercial microwave, and a commercial mini fridge.

5.3 EAST AND WEST WINGS

5.3.1 DASH RESTROOMS

The Dash is the connection point between the Wing Buildings and the Main Building. Both the West Wing and East Wing have their own Dash spaces, containing various MEP spaces, vertical transportation, and some staff and collection spaces. The restrooms on the Second and Fourth Floor of the Wing Dash Spaces have direct access to the Main Building, so they were designed to accommodate public use. These restrooms will be renovated to accommodate (3) Gender Neutral Single Use restrooms.

5.0 ARCHITECTURE

5.3.2 WING RESTROOMS

The Wing restrooms are non-public male and female restrooms, with a custodial closet. These facilities will be brought to equivalent square footages to provide universal accessibility, as well as efficient sanitary stacks. When the women's restrooms in the wings were originally built, they each were adjacent to a vestibule space. Over time, some of these spaces have become programming for the needs of the department occupying the floor. We have provided restroom layouts to accommodate the sanitary and restroom layout design around these department's programmatic needs. Further coordination to accommodate these ancillary programs will occur in later phases.

5.3.3 EAST COURT FAMILY RESTROOM

The East Court Second Floor contains existing public restrooms including a family restroom. While the existing family restroom meets some accessibility requirements, the design intent is to update the space to improve the universal accessibility by providing improved clearances, fixtures, accessories including a new adult changing table. This is the second Family Restroom containing a new adult changing table as part of this project.

5.4 PHASING

The project is divided into three primary phases:

Phase 1 – Main Building

Phase 2 – West Wing

Phase 3 – East Wing

Each phase will be executed using a top-down construction approach, beginning in the attic, and progressing downward. This methodology is driven by the requirements of the piping replacement strategy, which is detailed further in the Plumbing section of this report.

5.4.1 Phase 1 – Main Building

Phase 1 is further subdivided into the following sequential subphases, organized by building sections:

Phase 01A – Mobilization

Phase 01B – West Hall

Phase 01C – Rotunda

Phase 01D – East Hall

Phase 01E – North Hall

Phase 01F – East Range / East Court

Phase 01G – West Range / West Court

5.0 ARCHITECTURE

These subphases are arranged based on the contractor's primary access point at the West Hall Dock, with work proceeding in a logical sequence according to the physical adjacency of each area. The contractor is required to fully complete each subphase before advancing to the next.

5.4.2 Phases 2 and 3 – West and East Wings

Phases 2 and 3 will follow a similar top-down and adjacent-based sequencing strategy, ensuring consistency in execution and minimizing disruption across the site.

6.0 INTERIOR DESIGN

This Interiors narrative has been updated to reflect the development of the NMNH Upgrade Sanitary and Storm design since the Implementation Study Phase. Please refer to the 100% Construction Documents drawings for additional information.

6.1 PROGRAM NARRATIVE

6.1.1 MAIN BUILDING

6.1.1.1 North Lobby Restrooms

The North Lobby Restrooms will receive new finishes, plumbing fixtures, and toilet accessories. This includes a new family restroom containing an adult changing table.

6.1.1.2 Third Floor Private Restroom

The Private Restroom will receive all new finishes, plumbing fixtures, and toilet accessories. Our intent is to meet universal accessibility requirements, and we will identify any conditions that may not meet these requirements.

6.1.1.3 Third Floor Women s Restrooms

The Women s Restrooms will receive all new finishes, plumbing fixtures, and toilet accessories. Our intent is to meet universal accessibility requirements, and we will identify any conditions that may not meet these requirements.

6.1.1.4 Third Floor Rose Room Pantry

The Rose Room Pantry will receive all new finishes. 6'-9" countertop w/ base and wall cabinets. Must meet accessibility requirements. Include undermount sink, microwave, UC fridge, and built-in trash receptacle. Patch and repair disrupted finishes in adjacent spaces. The existing walls in the Rose Room to receive new wall paint.

6.1.1.5 Third Floor Shared Office Pantry

The Shared Office Pantry will receive all new finishes. 6'-9" countertop w/ base and wall cabinets. Must meet accessibility requirements. Include undermount sink, microwave, UC fridge, and built-in trash receptacle.

6.1.2 EAST WING

6.1.2.1 Restrooms

The East Wing Restrooms will receive all new finishes, plumbing fixtures, and toilet accessories. Must meet accessibility requirements unless noted otherwise.

6.0 INTERIOR DESIGN

6.1.3 WEST WING

6.1.3.1 Restrooms

The West Wing Restrooms will receive all new finishes, plumbing fixtures, and toilet accessories. Must meet accessibility requirements unless noted otherwise.

6.1.4 EAST COURT

6.1.4.1 Family Restroom

The East Court Family Restroom will receive all new finishes, plumbing fixtures, and toilet accessories. This will also include an adult changing table. Must meet accessibility requirements unless noted otherwise.

6.1.5 TEMPORARY CONSTRUCTION

6.1.5.1 Construction Walls

Temporary construction walls to receive temporary finishes to provide a finished appearance while construction is underway. Finishes to include wall base, paint, and graphics. Graphics to be coordinated and provided by others.

6.2 INTERIOR ARCHITECTURAL MATERIALS GENERAL

Interior partitions will be identified as one of the following types:

- 6 or 8 CMU braced above the ceiling to the structure above, located at perimeter walls of the restroom where indicated,
- 3 5/8 Cold Formed Metal Framing (CFMF): braced above the ceiling to the structure above, located at walls shared between spaces where indicated.
- Cold Formed Metal Framing (CFMF): braced above the ceiling to the structure above, located at furring conditions against existing walls and existing structure where indicated.

New doors will be painted hollow metal with painted hollow metal frames.

6.3 INTERIOR FINISHES

All materials specified shall be in accordance with all prevailing codes, regulations, and industry testing standards.

For the interior environment, materials shall be selected to reinforce the overlaying need for function, durability, and maintainability.

6.0 INTERIOR DESIGN

6.3.1 MAIN BUILDING

All toilet rooms in the Main Building will receive new finishes, plumbing fixtures, and toilet accessories. Pantries will receive new finishes and appliances. Patch and repair disrupted finishes in adjacent spaces. Provide temporary construction walls where required.

6.3.1.1 North Hall Restrooms

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Countertops: Solid Surface w/ solid surface angled panel apron, and undermount sinks
- Mirrors: Frameless w/ Wall mounted Lighting
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.
- Toilet Partitions: Phenolic floor mounted, overhead braced, No Sightline System

6.3.1.2 Third Floor Private Restroom

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Mirrors: Framed Mirror w/ wall mounted lighting.
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.

6.3.1.3 Third Floor Women s Restrooms

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Countertops: Solid Surface w/ solid surface angled panel apron, and undermount sinks.
- Mirrors: Frameless w/ wall mounted lighting.
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.
- Toilet Partitions: Phenolic floor mounted, overhead braced, No Sightline System.

6.0 INTERIOR DESIGN

6.3.1.4 Third Floor Rose Room Pantry

- Floor: Luxury Vinyl Tile (Planks) Patch & Repair adjacent carpet, Provide resilient floor transition
- Ceiling: N/A
- Walls: Painted Drywall -Existing walls in Rose Room to receive new wall paint to match.
- Base: To match existing
- Countertops: Solid Surface w/ integral backsplash, waterfall edge where indicated on drawings.
- Cabinets (Base & Wall): Plastic Laminate

6.3.1.5 Third Floor Shared Office Pantry

- Floor: Luxury Vinyl Tile (Planks)
- Ceiling: Painted drywall soffit.
- Walls: Painted to match existing adjacent finish.
- Base: To match existing adjacent finish.
- Countertops: Solid Surface w/ integral backsplash
- Cabinets (Base & Wall): Plastic Laminate

6.3.2 EAST WING

All toilet rooms in the East Wing will receive new finishes, plumbing fixtures, and toilet accessories. Provide temporary construction walls where required.

6.3.2.1 Gender Neutral/Single Use Restrooms

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Countertops: Solid Surface w/ solid surface angled panel apron, undermount sinks
- Mirrors: Frameless w/ Wall mounted Lighting
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.

6.3.2.2 Restrooms

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Countertops: Solid Surface w/ solid surface angled panel apron, undermount sinks
- Mirrors: Frameless w/ Wall mounted Lighting
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.
- Toilet Partitions: Phenolic ceiling mounted, No Sightline System

6.0 INTERIOR DESIGN

6.3.3 WEST WING

All toilet rooms in the West Wing will receive new finishes, plumbing fixtures, and toilet accessories. Provide temporary construction walls where required.

6.3.3.1 Gender Neutral/Single Use Restrooms

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Countertops: Solid Surface w/ solid surface angled panel apron, undermount sinks
- Mirrors: Frameless w/ Wall mounted Lighting
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.

6.3.4 RESTROOMS

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Countertops: Solid Surface w/ solid surface angled panel apron, undermount sinks
- Mirrors: Frameless w/ Wall mounted Lighting
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.
- Toilet Partitions: Phenolic ceiling mounted, No Sightline System

6.3.5 EAST COURT – FAMILY RESTROOM

The Family Restroom on the East Court, Second Floor, will receive new finishes, plumbing fixtures, and toilet accessories. Patch and repair disrupted finishes in adjacent spaces. Provide temporary construction walls where required.

- Floor: Large Format Ceramic Tile
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Full-height Ceramic Wall Tile, including metal edge trim at outside corners
- Base: Metal Cove Base
- Mirrors: Framed Mirror w/ wall mounted lighting.
- Toilet Accessories: Provide all new Accessories and Waste Receptacles as indicated.

6.3.6 WEST WING - THIRD FLOOR COPY ROOM

- Floor: Luxury Vinyl Tile (Planks)
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Painted Drywall

6.0 INTERIOR DESIGN

- Base: Rubber base

6.3.7 EAST AND WEST WING - FOURTH FLOOR COPY/MICROWAVE ROOM

- Floor: Luxury Vinyl Tile (Planks)
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Painted Drywall
- Base: Rubber base

6.3.8 WEST WING - FIFTH FLOOR COPY/MICROWAVE ROOM

- Floor: Luxury Vinyl Tile (Planks)
- Ceiling: Acoustic Ceiling Tile (24x48)
- Walls: Painted Drywall
- Base: Rubber base

6.3.9 CUSTODIAL CLOSETS

- Floor: Sealed Concrete
- Ceiling: No Ceiling - Exposed
- Walls: Epoxy Paint w/ Wall Protection Panels (4 -0 AFF)
- Base: Rubber base

6.3.10 CONSTRUCTION WALLS

- Walls: Paint w/ Graphics to be Coordinated and Provided by others
- Base: Rubber base

6.4 TOILET ACCESSORIES

6.4.1 RESTROOMS

All restrooms will receive the following toilet accessories:

Refer to standard toilet room mounting height sheet for toilet accessory requirements per stall.

- (1) TA-01 surface toilet tissue dispenser roll type per toilet
- TA-05 recessed hand dryer – at least (1) per restroom, see drawings for exact counts.
- (1) TA-07 surface mounted waste receptacle per restroom
- (1) TA-08 surface napkin / tampon vendor per women's restroom
- (1) TA-10 surface napkin disposal per toilet in each women's restroom
- TA-13 surface soap dispenser – at least (1) per restroom, see drawings for exact counts.
- (1) TA-16 surface seat cover dispenser per toilet
- (1) TA-19 mop strip per custodial closet
- (1) TA-20 framed mirror where indicated over wall mounted lavatories only
- (1) TA-23 surface baby changing station horizontal see drawings for locations
- TA-28 grab bar straight horizontal as required for accessible and ambulatory toilets.
- TA-29 grab bar straight vertical as required for accessible and ambulatory toilets.

6.0 INTERIOR DESIGN

- Frameless mirrors 4'-0" high over length of counters see elevations for widths.
- Full-length mirror – at least (1) per toilet room

6.4.2 PANTRIES

Both Pantry areas will receive the following accessories:

- (1) TA-03 surface paper towel dispenser
- (1) TA-13 surface soap dispenser

7.0 STRUCTURAL

7.1 APPLICABLE CODES AND STANDARDS

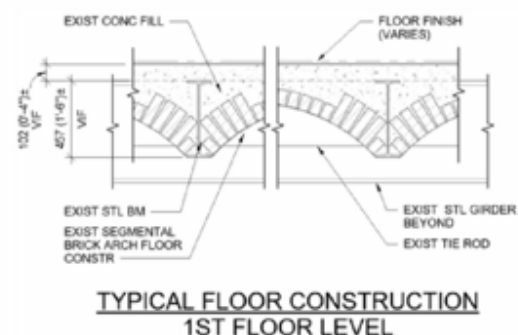
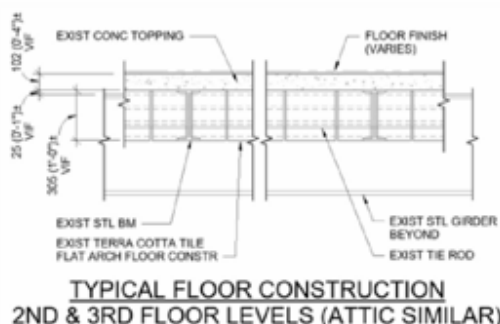
- IBC 2021 ICC International Building Code
- IEBC 2021 ICC International Existing Building Code
- IEBC 2021 International Building Code
- ACI 318-19 American Concrete Institute - Building Code for Structural Concrete
- ASCE 7-16 American Society of Civil Engineers - Minimum Design Loads and Associated Criteria for Buildings and Other Structures
- AISC 360-16 American Institute of Steel Construction - Specification for Structural Steel Buildings
- ASTM - American Society for Testing and Materials
- AWS D1.1 American Welding Society - Structural Welding Code for Steel
- TMS 402-16 The Masonry Society - Building Code Requirements for Masonry Structures

7.2 EXISTING CONDITIONS SUMMARY

The Natural History Museum structure consists of additions, renovations, and infills which have added to or modified the original building structural systems over its 100 plus year lifespan. The original main building consisted of a partial basement and three supported floors in three halls and two range spaces. The original structure was constructed with heavy masonry walls and piers bearing on unreinforced concrete spread footings. The floors consist of riveted steel plate columns and girders bearing on masonry piers. The steel structure supports segmental brick arch and concrete floors at the first-floor level and terracotta flat arches with concrete topping at the upper floors. The roof structure is a cast-in-place concrete one-way slab supported by reinforced concrete rafters and riveted steel trusses.

The east and west wing additions were added in 1960 and 1963 respectively as independent structural systems isolated from the main building with expansions joints. The building additions each consist of full basements and six supported floors constructed from cast-in-place concrete foundations and columns supporting concrete flat plate floors with drop panels at column locations.

Both main building courtyards have been infilled with structures tied into the original building walls and framing. Additional interstitial floor mezzanines have also been added over the years. These infill areas and mezzanines consist of a variety of different structural systems which can vary from steel or concrete framing with concrete slabs to wood or gage metal bearing walls and floors.



7.0 STRUCTURAL

7.3 DESIGN CRITERIA

7.3.1 RISK CATEGORY

IBC 2021 Table 1604.5

Risk Category III

7.3.2 DESIGN LIVE LOADS FOR STRUCTURE SUPPORTING NEW WORK

Public Rooms

100 psf

Corridors Serving Public Rooms

100 psf

7.3.3 SEISMIC LOAD CRITERIA

Seismic Importance Factor

$I_E = 1.25$

0.2 Sec Spectral Response Acceleration

$S_S = 0.133$

1.0 Sec Spectral Response Acceleration

$S_1 = 0.043$

5% Damped Spectral Response Acceleration at Short Periods

$S_{DS} = 0.142$

5% Damped Spectral Response Acceleration at 1-Second Period

$S_{D1} = 0.069$

Seismic Design Category

B

7.3.4 MATERIALS FOR NEW WORK

Steel Wide Flange Sections, Channels, Angles

ASTM A992 Gr. 50

Steel Plates

ASTM A36 Gr. 36

7.4 STRUCTURAL SYSTEMS

7.4.1 STORM DRAINAGE ROUTING

In general, relocated storm drains will be rerouted horizontally to areas with vertical distribution of utilities where existing penetrations or shaft openings through floor systems can be used. The horizontal routing of piping to these areas may require penetrations through existing clay tile or CMU walls. It is anticipated that these wall penetrations will be small enough to not require reinforcement of the walls around these penetrations. However, if any area of clay tile wall is determined to be too unstable for a penetration, it may be necessary to locally demolish sections of wall and replace with a gypsum on cold formed metal framing or CMU wall in the same location.

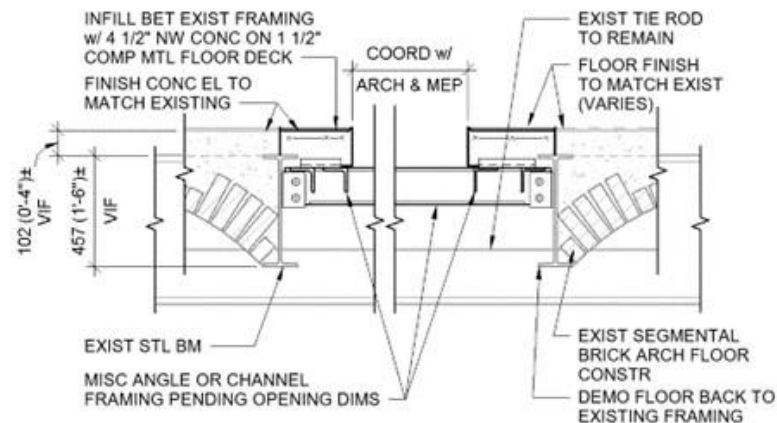
7.4.2 RESTROOM AND PANTRY RENOVATIONS

The renovation of the existing restrooms will improve or keep the number of fixtures the same as existing, which will allow the reuse of the existing utility routing. It is anticipated that there are limited locations where restroom and pantry layouts will require relocated chase areas and additional or relocated floor drains. These special conditions will require some additional work at the floor penetrations.

7.4.3 FLOOR PENETRATIONS THROUGH EXISTING FLOOR ASSEMBLIES

Where proposed work cannot utilize existing floor openings and requires a penetration through the existing floor, careful coordination with existing floor assemblies will be required.

7.0 STRUCTURAL



PROPOSED OPENING 1ST FLOOR LEVEL

If the work occurs in an area of the original segmental brick arch construction or terra cotta flat arches, steel frames consisting of angle or channel members may be required around the proposed openings. Frames must be installed, spanning between existing steel beams, prior to cutting penetrations to support the areas of the brick arch system which will be disturbed. Openings must be coordinated to avoid tie rod locations. Tie rods must remain unless the complete floor system is removed and replaced as described below. Larger openings or groups of smaller penetrations may require a section of the floor system to be removed entirely between main framing members and replaced with a concrete slab on metal deck.

7.4.4 ACCESS TO SERVICE TUNNELS

Existing hatches into many of the service tunnels are too small or congested to allow reasonable access to areas required to complete the proposed work. These hatches will either be enlarged, or new hatches will be added in areas where work will need to be completed. The structural cover on the existing tunnels is an approximately 9 thick concrete slabs reinforced with expanded steel mesh. Angle frames will be installed on all sides of new or enlarged openings and anchored into the underside of the existing slab.

7.4.5 UTILITY AND PARTITION SUPPORT FRAMING

In areas of work within the Main Building where floors are comprised of brick or terra cotta assemblies, as described in Section 8.2, MEP and Architectural systems are not permitted to be supported by or braced to the floor assembly. Connections must only be made to the steel framing members. This work will require the local removal of plaster ceilings to allow steel hanger tabs to be welded onto the existing beam's bottom flange. All exposed steel must be covered after installation by spray fire resistant material or intumescent coatings at the architect's direction to restore the existing floor's fire rating.

7.0 STRUCTURAL

In many areas, the existing framing may not be located close enough to where supports are required. At these locations, supplemental framing between existing beams or slotted channel framing members will be required. In areas with particularly dense MEP and partition requirements it may be preferred to install a suspended grid of slotted channel framing in the whole space to allow more flexibility in the layout of the supported systems.

8.0 MECHANICAL

8.1 APPLICABLE CODES AND STANDARDS

The following codes and guidelines have been referenced.

- International Building Code (IBC) 2021 Edition
- International Existing Building Code (IEBC) 2021 Edition
- International Mechanical Code (IMC) 2021 Edition
- International Energy Conservation Code (IECC) 2021 Edition
- ASHRAE standard 62.1 2022 Edition
- ASHRAE standard 90.1 2022 Edition
- ASHRAE HVAC Fundamentals Handbook
- ASHRAE HVAC Applications Handbook
- ASHRAE HVAC Systems and Equipment Handbook
- NFPA -All applicable codes and standards
- Energy Policy Act (EPACT)
- SMACNA Duct Standards - All applicable standards (including duct leakage)
- (ASME) American Society of Mechanical Engineers -All related documents
- (ASTM) American Society for Testing and Materials International -All related documents
- (ANSI) American National Standards Institute -All related documents
- Smithsonian Standards and Guidelines, including:
 - Fire Protection and Life Safety Design Manual
 - Smithsonian Facilities Design Standards

8.2 EXISTING CONDITIONS

The West Wing Southwest bathrooms are served by a common exhaust shaft for all floors up to toilet exhaust fan E-40 in the penthouse. The West Wing Northeast bathrooms are also served by a common exhaust shaft up to toilet exhaust fan E-48 in the penthouse. The Northeast bathrooms on the fourth and sixth floors have exterior exposure and are provided with hot water finned tube heating. The West Wing ground, first, and second floor have been renovated within the last 15 years and bathrooms are currently provided with supply air from terminal units.

The East Wing Northwest bathrooms are served by a common exhaust shaft for all floors up to toilet exhaust fan E-15 located in the penthouse. The northwest bathrooms on the second, fourth, and sixth floors have exterior exposure and are provided with hot water-finned tube heating. The East Wing Southeast bathrooms are served by a common exhaust shaft up to toilet exhaust fan E-17 located in the penthouse.

Large supply and return duct mains run throughout the attic. Branch ducts with terminal equipment serve halls below. Existing storm piping is located next to ductwork mains and above or below branch ductwork.

8.0 MECHANICAL

8.3 DESIGN CRITERIA

8.3.1 CLIMATE CONDITIONS

Site Location	Washington, District of Columbia 38.892° Lat., 77.03° Long., 6.6 ft. (2 m) Elevation
Climatic Location	Reagan International Airport, VA 38.847° Lat., 77.035° Long., 10 ft. (3 m) Elevation
Climate Zone	4A
<u>Summer</u>	
Dry Bulb Temp ¹	94.5° F (34.7 ° C)
Coincident Wet Bulb ¹	75.6° F (24.2 ° C)
Enthalpy ¹	38.986 BTU/lb (90.68 kJ/kg)
Evaporative Wet Bulb ²	78.4° F (25.8 ° C)
Coincident Dry Bulb ²	89.1° F (31.7 ° C)
Enthalpy ²	41.878 BTU/lb (97.41 kJ/kg)
Dehumidification DP ³	6° F (24.2 ° C)
Coincident Dry Bulb ³	83.3° F (28.5 ° C)
Enthalpy ³	
<u>Winter</u>	
Dry Bulb Temperature ⁴	41.098 BTU/lb (95.59 kJ/kg) 17.1° F (-8.3 ° C)

¹Source: 2021 ASHRAE Fundamentals 0.4% Cooling Design Condition for peak dry bulb and mean coincident wet bulb.

²Source: 2021 ASHRAE Fundamentals 0.4% Cooling Design Condition for peak wet bulb and mean coincident dry bulb.

³Source: 2021 ASHRAE Fundamentals 0.4% Cooling Design Condition.

⁴Source: 2021 ASHRAE Fundamentals 99.6% Heating Design Condition.

8.3.2 VENTILATION

Exhaust air quantities for the spaces will be in accordance with ASHRAE 62.1 2022 Ventilation for Acceptable Indoor Air Quality.

8.3.3 INDOOR DESIGN CRITERIA

Room conditions will be designed for a minimum of 68°F +/- 1.5 F (20°C +/- .825°C).

8.4 HVAC LOADS

All load calculations will be performed utilizing state-of-the-art hourly energy load computer simulations (Carrier HAP v5.11 software). Computer simulation software will calculate loads in accordance with ASHRAE Standard 183. Reference Appendix A for hand load calculations.

Lighting loads are assumed to be compliant with previous ASHRAE 90.1 minimum lighting power densities. Safety factors of 10% for heating and cooling are used to account for variance.

8.0 MECHANICAL

Units are sized according to worst-case outside air conditions based on enthalpy, which is evaporative 0.4%.

8.5 COOLING AND HEATING PLANTS

Existing hot water heating plant will be utilized for new equipment to the maximum extent possible. The hot water entering temperature is 180° F (82.2°C).

8.6 HVAC SYSTEMS

For each renovated bathroom, the branch exhaust and supply ductwork will be removed back to the tie in point at the existing main and existing air devices will be removed. Pre-Demolition airflow measurements will be made on each floor to confirm existing airflows. New branch ductwork will be installed and connected to new air devices within the bathroom as needed. Exhaust airflows will be increased as needed to comply with updated code minimum exhaust rates. Existing fans will be rebalanced as needed for change in airflow. Existing finned tube will be demolished and replaced with new electric radiant panels in bathrooms with exterior exposure. Hot water piping will be altered as needed.

Ductwork, piping, and air devices will need to be relocated and/or replaced as needed for new storm piping installation. Areas of particular concern for ductwork coordination are the Northwest, Northeast, and North side of West Hall sections of the Attic. These locations have a significant amount of ductwork and little space to relocate. Refer to the 100% Construction Documents drawings for the extent of duct rerouting required. See pictures of example locations below:

Branch duct above storm in Northwest attic (Northeast similar)



8.0 MECHANICAL

West Hall North side branch ductwork in path of extended storm



West Hall South side branch ductwork in path of new storm



East Hall South Side Duct Conflicts with rerouted storm



8.0 MECHANICAL

8.7 HVAC CONTROLS

All new equipment requiring controls will connect to the existing SI BAS system through BACnet.

8.8 COMMISSIONING

All existing equipment altered, relocated, or replaced shall be pre-commissioned to determine pre-demolition airflow and proper operation. After the final construction phase is complete and the equipment is returned to normal operation, the system shall be recommissioned.

9.0 PLUMBING

9.1 APPLICABLE CODES AND STANDARDS

The latest edition of the following codes and standards shall be used, except where a specific version is listed. Where a deviation or conflict occurs, the more stringent code or standard shall be applied.

- Smithsonian Institution Facilities Design Standards - 2021 Edition.
- International Plumbing Code (IPC) - 2021 edition
- International Energy Conservation Code (IECC) - 2021 edition
- International Building Code (IBC) 2021 Edition
- International Existing Building Code (IEBC) 2021 Edition
- International Energy Conservation Code (IECC) 2021 Edition
- ASHRAE standard 90.1 2022 Edition
- NFPA - All applicable codes and standards
- Energy Policy Act (EPACT)
- (ASTM) American Society for Testing and Materials International - All related documents
- (ANSI) American National Standards Institute - All related documents
- Smithsonian Institution, Office of Facilities and Operations, Codes, Standards and Guidelines - Latest edition
- Smithsonian Sustainability Plans - Latest edition
- 10 CFR 434 - Energy Code for New Federal Commercial and Multi-Family High-Rise Residential Buildings. Plumbing Drainage Institute Standard PDI-WH-201, Appendix A, Domestic Water System Sizing.
- DC Water and Sewer Authority (DCWASA) Requirements

9.2 EXISTING CONDITIONS ASSESSMENT (IMPLEMENTATION PHASE)

The Design Team was tasked with providing an extensive implementation study to determine the conditions of the existing sanitary, vent and storm systems that serve the NHB. Over the course of two years, it was discovered the systems had in most cases exceeded their life expectancy and action would be necessary to upgrade those systems.

Although there are areas that have been renovated over the years and have had the sanitary and vent systems modified or replaced, there are storm piping systems that have been left untouched behind walls in sensitive areas that need to be addressed as well as sanitary and vent piping systems that serve outdated men's and women's rooms that no longer meet accessibility standards or the current desires of the building.

Below is the current design intent to meet these requirements and provide an understanding of both the extent and complexity of the proposed project.

9.0 PLUMBING

9.3 DESIGN CRITERIA

9.3.1 STORM DRAINAGE SYSTEMS

New storm piping will be used to replace existing storm pipes as required to upgrade the building's storm system. The existing storm system routing will be modified to avoid sensitive storage areas, exhibits and other areas of concern. Phasing of the storm piping installation will be coordinated with areas of concern as well as the weather and time of year. Existing storm systems that will not need to be rerouted and where pipes are in open areas or areas of less concern will be replaced in kind as much as possible to minimize impact to spaces.

All storm drainage piping above grade within the building will be cast iron in accordance with SI Standards. Piping in attic spaces where indicated shall be schedule 80 PVC.

The new storm drainage piping will discharge to the existing storm system from the newly located stack locations and extend horizontally to the existing tunnels if approved to be re-used. If not approved, all new storm drainage piping will discharge to the site storm sewer by gravity through new mains to be installed in the tunnels.

In the Main Building, new storm piping will be designed to extend across the attic floors to new stack locations to reduce or eliminate the current concerns of compromising any sensitive areas. The extent of the pipe run and routing on the attic floors will be determined by assessing the possible pitches associated with the storm pipe and the existing structural steel. Storm pipe runs may need to be shortened and therefore penetrate the attic floor to the ceiling space below where new soffits will be provided to accommodate the pipe routing to the newly proposed storm stack locations.

Any proposed abandoned storm piping systems will have all open ends capped and the entire cavity filled with an approved Class A fire-rated material to safeguard the system from any future reconnections to the abandoned systems.

Horizontal storm drainage piping aboveground will be insulated with fiberglass insulation. All horizontal and vertical Insulation in exposed areas will be covered with a sealed vinyl jacket.

Due to the rising frequency of severe storms on the East Coast, the building storm drainage system shall be sized in accordance with tables and flow rates as stated on the IPC - International Plumbing Code - 2021 for a 3.5" / hr. rainfall rate. This will slightly oversize the piping system to reduce potential overflow conditions in the building.

9.0 PLUMBING

9.3.2 SANITARY WASTE AND VENT

New sanitary and vent systems will be provided as required to serve the proposed restroom modifications, pantry additions and to satisfy the staff and public accessibility requirements.

The phasing of the sanitary replacement systems and restroom renovations will be coordinated with the staff, facilities, and historic preservation personnel to ensure there is minimal disruption to the everyday operations of the museum.

The new sanitary systems will extend through the building and into the access tunnels to connect to existing systems that may have been deemed suitable to be re-used and scheduled to remain. If the sanitary main is not in satisfactory condition, a new sanitary main will be provided and extended horizontally in the tunnels to the existing service locations exiting the building to the site systems.

All sanitary waste and vent piping shall be located under floor or building slabs with vertical risers located in existing or new pipe chases as required. All cleanouts in the drainage system shall be accessible.

Similar to the storm system, any proposed abandoned sanitary piping systems will have all open ends capped and the entire cavity filled with an approved material to safeguard the system from any future reconnections to the abandoned systems.

The system will be designed to maintain a minimum velocity of two feet per second. Minimum pipe slopes will be 2% for 2.5-inch and smaller piping, 1% for 3-inch and larger piping.

Floor drains shall be provided for common area restrooms with more than three plumbing fixtures. Drains and traps that are subject to evaporation shall be provided with trap primers or inline trap guards if accepted by NMNH facilities and the local Washington DC AHJ.

The sanitary drainage flow rates and pipe sizes shall be based on drainage fixture unit values of the actual fixtures and appropriate diversity factors and allowances.

9.3.3 POTABLE COLD WATER (RENOVATED RESTROOMS AND NEW PANTRIES)

The new potable cold-water system scheduled to serve the renovated restrooms and pantries will be extended with new supply risers from the existing supply main located in the basement of each building. The new cold-water system will be provided with isolation valves to each area served.

The potable cold-water piping shall be sized to maintain a minimum pressure of 30 psig at the most remote fixture.

Cold water will be extended to serve all fixtures requiring cold water including hose bibbs located below public lavatory counters.

9.0 PLUMBING

Water velocity in the cold-water distribution piping shall not exceed 8 feet per second and provisions shall be made to arrest water hammer. System pipe sizes for cold water shall be based on fixture unit values with appropriate diversity factors and actual equipment demands.

The new cold water piping system for these areas shall be insulated with ½-inch mineral fiberglass insulation. The domestic water system shall be sterilized with hypo chloride solution.

9.3.4 POTABLE HOT WATER (RENOVATED RESTROOMS AND NEW PANTRIES)

The new potable hot water system scheduled to serve the proposed restrooms and pantries will be extended from the existing hot water supply that is currently serving the areas plumbing fixtures. The new hot water system will be provided with isolation valves to each area served. Mixing valves will be provided at each terminal device.

New Hot water return piping, and pumps will be as provided to recirculate the hot water system back to the active generating hot water source in both East and West Wings stacked restrooms.

Point of use type water heaters will be provided at the new public lavatories and pantry sinks. A new storage tank type water heater will be provided to replace the existing unit at the West Wing Loading dock.

The potable hot water piping shall be sized to maintain a minimum pressure 30 psig at the hydraulically most remote plumbing fixture. Water velocity in the hot water distribution piping shall not exceed 8 feet per second. System pipe sizes for the hot water piping system will be based on fixture unit values with appropriate diversity factors and actual equipment demands.

The new hot water piping system for these areas will be insulated with 1-inch mineral fiber or elastomer type insulation. The domestic water piping system will be sterilized with hypo chloride solution.

9.3.5 EXISTING OFFICE SINK LOCATIONS

The museum currently contains offices with kitchen sinks, lab sinks, or wall hung type lavatories. The existing sinks at these locations are to remain in all locations except on the Third Floor of the Main Building. These sinks will be removed and all associated waste and water piping back to a point suitable for capping.

9.0 PLUMBING

9.3.6 PLUMBING FIXTURES AND SPECIALTIES

9.3.6.1 Water Closets

Water closets will be Kohler, wall-hung or floor mounted 1.28 GPF flush valve type with Sloan exposed hard-wired sensor operated flush valves. Seats will be open front, less cover. Barrier-free universally accessible use units will have high bowl units with seats mounted 17"-19" above the floor.

9.3.6.2 Water Closets – Tank Type

Water closets will be Kohler, floor mounted, floor outlet 1.28 GPF tank type with Sloan exposed hard-wired sensor operated flush valves. Seats will be open front, less cover. ADA height.

9.3.6.3 Lavatories: Kohler

Countertop type integral bowl vitreous China units equipped with Sloan hard-wired sensor operated faucets with 0.50 GPM restricted flow.

9.3.6.4 Lavatories: Kohler

Wall-hung, ACCESSIBILITY height, vitreous China units equipped with Sloan hard-wired sensor operated faucets with 0.50 GPM restricted flow.

9.3.6.5 Urinals

Urinals will be Kohler, wall-hung, with Sloan hard-wired sensor operated flush valves, maintaining .0125 GPF (pint per flush) water saving.

9.3.6.6 Pantry Sinks

Elkay countertop-type undermount units, in 20-gauge stainless steel, with 1.5 to 1.75 GPM flow-restricted single lever faucets with pull-out spray, will be barrier-free universally accessible use units. These will be 5 deep sink bowl units with offset drain assemblies.

9.3.6.7 Mop Receptors

Molded stone type with hose end service sink faucet, complete with vacuum breaker, accessory mounting bracket, and hose.

9.3.6.8 Hose Bibb Interior

Chrome plated, wall mounted, hose bib with vacuum breaker, hose outlet.

9.3.6.9 Water Hammer Arrestors

Shall be provided where needed to reduce hydrostatic shock.

9.3.6.10 Backflow Preventers

Coffee makers/ice machines.

9.0 PLUMBING

9.3.6.11 Electric Water Coolers / Drinking Fountains

Stainless steel, barrier-free and universally accessible use.

9.3.6.12 Electric Water Heaters

Refer to electric water heater schedule on drawing P-501.

9.4 GENERAL

In addition to utilizing the existing tunnel access hatches on the ground level floor to perform the piping replacement, new or modified tunnel access hatches will need to be provided.

Due to the congestion in some of the tunnels, standard pipe lengths may need to be modified to permit the pipe to enter the tunnels.

Asset identification will be incorporated as part of the project and will utilize the Smithsonian Office of Facilities Management and Reliability Asset Hierarchy and follow the CSI Master Format System to the Equipment Type Level Guidelines 2023.

All pressure services shall be provided with branch isolation valves for each room or area they supply.

Sleeves shall be provided for all piping penetrations at walls and floors.

Fire stopping shall be provided for all piping penetrations at walls and floors. The fire stopping materials shall be tested to meet ASTM E814 standard. Fire stop material shall not contain refractory ceramic fibers and shall meet the requirements of NFPA 101.

Identification shall be provided for all piping systems, valves, equipment, etc. except as specified herein or where in conflict with the university standards. Marker and color legend shall comply with ANSI and OSHA standards.

9.0 PLUMBING

9.5 MATERIALS OF CONSTRUCTION

Listed below are the materials that will be provided and specified.

9.5.1 PIPING SPECIFICATIONS

9.5.1.1 Sanitary and Storm Water Below Grade

Polyvinyl Chloride DWV piping with solvent weld joints. (Approval to be verified with NMNH Facilities)

9.5.1.2 Storm Water Main Building (Attics Only)

Schedule 80 PVC piping with solvent weld joints.

9.5.1.3 Sanitary and Storm Water Above Grade or in Tunnels

Cast iron. No-hub with 4-band mechanical couplings.

9.5.1.4 Insulation

Fiberglass on horizontal piping and underside of drains within the building.

9.5.1.5 Potable Water

Type L copper with lead-free soldered joints.

9.5.1.6 Cold and Hot Water Insulation

Fiberglass.

9.5.1.7 Valves

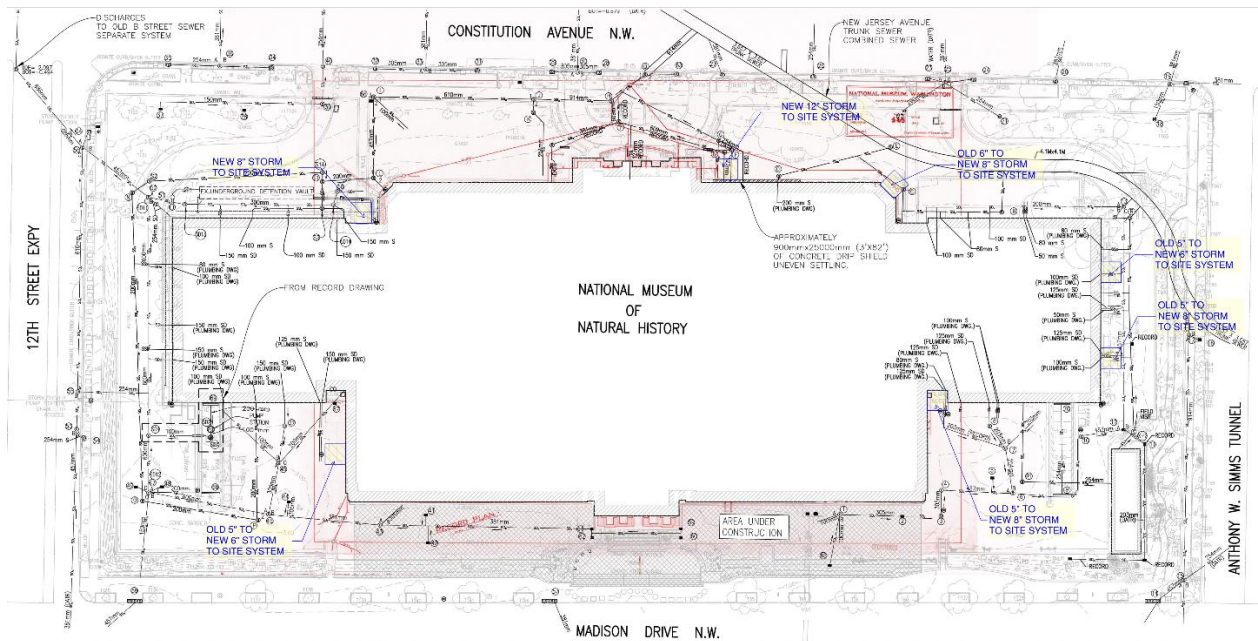
Bronze ball valves and epoxy-coated iron ball valves.

9.0 PLUMBING

9.6 PLUMBING AND UTILITY INTERFACE

The new storm water systems have been sized per the IBC 2021 storm drainage sections and have impacted the existing pipe sizes exiting the building. Some existing locations will require existing pipe cores / sleeves to be enlarged while other locations will require new penetrations. Basement and tunnel locations requiring new or larger core penetrations will be provided with a new Link Seal at each location. Exterior storm piping systems will be coordinated with existing and new storm piping exiting the building to determine the proposed utility connection locations.

Site plan indicating (7) new pipe penetration locations. Locations shown in Blue, 1900's utility overlay in Red.



10.0 ELECTRICAL

10.1 APPLICABLE CODES AND STANDARDS

The latest edition of the following codes and standards shall be used, except where a specific version is listed. Where a deviation or conflict occurs, the more stringent code or standard shall be applied.

- International Building Code (IBC), 2021 Edition
- NFPA 70, National Electrical Code (NEC), 2021 Edition
- International Energy Conservation Code (IECC), 2021 Edition
- NFPA 101, Life Safety Code, 2024 Edition
- National Electrical Safety Code, ASNI C2
- Underwriters Laboratories, Inc. (UL)
- National Electrical Manufacturers Association (NEMA)
- American National Standards Institution (ANSI)
- Factory Mutual, Inc. (FM)
- Smithsonian Institution Office of Safety, Health, and Environmental Management: Fire Protection & Life Safety Design Manual
- Local City Regulations

The following standards shall be included in the electrical system design, in addition to requirements of the Smithsonian Institution Design Standards document, as issued by the Office of Engineering Design and Construction.

- All electrical wiring shall be in conduit and concealed where possible. Flexible cables will only be permitted for 6-foot whips to luminaires.
- Receptacle wiring: Type THHN/THWN conductors. Minimum wire size 12 AWG. Separate neutrals will be provided for each circuit. Common neutrals for multiple phases will not be permitted.
- All branch circuit wiring will be sized to keep overall voltage drop to a maximum of 3%.
- Duplex receptacles will be single phase, 120V, grounding type or ground fault type as required, NEMA 5-20R, flush-mounted.
- Each receptacle will be identified with a panelboard and branch circuit number.

10.2 EXISTING CONDITIONS

The Main Building, East Wing, and West Wing buildings each have their own set of network transformers supplying dedicated switchboards in the buildings respective transformer vaults supporting distribution throughout the buildings. The electrical panels downstream that are currently feeding the existing bathrooms and new pantry areas will be utilized.

10.3 NEW WORK

New electrical work will be in support of renovated bathrooms and new pantries in addition to any electrical modifications required because of the new sanitary and storm systems. New lighting, electrical devices, plumbing equipment & mechanical equipment will be circuited to the nearest existing panel. Power will also be provided with new instantaneous point-of-use water heaters throughout the buildings. Spare circuit breakers will be utilized where possible. All new circuit breakers will match the existing panel short circuit rating. All new lighting will be energy-efficient LED with automatic controls to meet all current codes and standards.

11.0 SUSTAINABILITY

11.1 APPLICABLE CODES AND STANDARDS

The National Museum of Natural History (NMNH) Upgrade Sanitary and Storm project will adhere to the Smithsonian Institution's commitment to sustainability. The project's sustainable focus points are as follows:

11.1.1 LEED

Although the project will not be LEED certified, it will follow the Smithsonian's sustainability guidelines, ensuring that the project is executed in an environmentally friendly manner.

11.1.2 LIFE CYCLE

The life cycle of a building supply refers to the stages that a material or product goes through, from its extraction and manufacturing to its use and final disposal. The project will prioritize the selection of products for piping, finishes, fixtures, and accessories that have a long life cycle. This approach reduces waste and the need for frequent replacements, contributing to sustainability.

11.1.3 WATER CONSUMPTION

The project aims to reduce water consumption by using water-reducing plumbing fixtures, known as low-flow fixtures. A low-flow plumbing fixture is a water-saving device designed to reduce water consumption by providing a lower flow rate or smaller quantity per flush. These fixtures include faucets, urinals, and toilets. This not only lowers the project's environmental footprint but also results in long-term cost savings. Refer to the Plumbing section for more information on fixture selections.

12.0 HAZMAT

12.1 HAZMAT

The NHB has been the subject of multiple historical evaluations for various hazardous materials including Asbestos Containing Materials (ACM), Lead Based Paint (LBP), and Lead Containing Paint (LCP); SI maintains a considerable number of these records as far back as the 1980s. Historical records, while very broad, have not been exceedingly cohesive or usable for specific projects.

As part of the Implementation Study Phase, F&R prepared a Preliminary Limited Regulated Survey Report (Draft 2) dated June 1, 2022, which documented evaluation efforts performed in May 2022. The report identified numerous suspect ACMs which were homogenized and sampled as well as paint films which were evaluated for lead content and resulted in a rough understanding of conditions at the site. Supplemental evaluation may be warranted as the development of the design proceeds and construction activities are more well defined. The Hazmat Team developed project specific hazmat guidance as part of the 100% Construction Documents Phase.

The Hazmat Report, included under a separate cover for information, is the result of the sampling and analysis work performed by Froehling & Robertson and the contract laboratory, Environmental Hazards Services, LLC.

13.0 CIVIL

13.1 CIVIL

13.1.1 Existing Site Conditions

The National Museum of Natural History (NMNH) is located in Washington, DC south of Constitution Ave, north of Madison Dr and between 12th St NW and 9th St Expressway. See Figure 1 below. The site is split between two major drainage systems in the District; 1. Municipal Separated Storm Sewer System (MS4) and 2. Combined Sewer System (CSS). The eastern half of the site drains to the B Street/New Jersey Trunk Sewer system, which is a combined sewer system. The western half of the site drains to a separate storm drain only system located in Constitution Avenue. Both systems drain to the Potomac River. The sanitary sewer for the entire site discharges to the New Jersey Trunk Sewer tunnel. The New Jersey Trunk Sewer is located under Constitution Avenue and flows southeast to the Blue Plains wastewater treatment plant and overflows into the Anacostia River. There are several existing connections between the building sanitary, and storm drain system to the public system. A portion of the building's roof drains are piped underground to a structural stormwater vault in the northwest, which discharges directly to the MS4 watershed. Many of these connections are not adequately sized to convey the discharge, thus as part of this project, the connections will be replaced and upsized in seven (7) locations.

13.1.2 Proposed Conditions

This project's focus is to upgrade the storm drain and sanitary sewer systems inside the building. As part of the upgrades, exterior connections/improvements to the existing storm drain and sewer systems on the museum property are required. There are seven locations that have been identified that require larger connections between the building and sanitary/storm drain systems.

The proposed exterior pipe sizing ranges between 6" and 12". The exterior pipe material shall be PVC schedule 40, unless otherwise determined by the Owner. Connections to the existing systems vary between existing structure/manhole modifications or field connections.

Contractor staging areas and storage areas have also been identified on the east side of the site.

Lastly, an erosion and sediment control plan has been prepared for permitting and Contractor use. The erosion and sediment control approach includes silt fence and utility trenching best practices. These best practices include minimizing earth disturbance, stabilizing disturbed earth as soon as possible, and maintaining a clean work area.

13.0 CIVIL

13.1.3 Permitting

The disturbed area to make the connections exceeds 50 sf and triggers the need for an erosion and sediment control plan and building-civil (B-Civ) permit. Erosion and Sediment Control plans, which reflect the scope of the work in addition to Contractor work and staging areas, will be reviewed by the District Department of Buildings (DOB) and the District Department of Energy and Environment (DOEE) for approval. Upon approval, the permit will be issued to the Smithsonian for their use. The DOB permits are valid for 1 year after issuance OR the 1 year after the most recent inspection, whichever is later. Permits can be extended for 6-month intervals within 30 days prior to expiration.

14.0 PROJECT TEAM

14.1 ARCHITECT/ENGINEER/INTERIOR DESIGNER

EwingCole

Project Manager	Roger Rudy
Project Architect	Troy Varner
Architecture Staff	Patrick Papia
Senior Architect	Ryan Delahoussaye
Interior Designer	Becca Kuhn
Plumbing Engineer	Jeff McLoughlin
Plumbing Staff	Angel Colon
Lead Mechanical Engineer	Matthew Hayes
Electrical Engineer	Jacob Falcon
Fire Protection Engineer	Roger Rudy
Structural Engineer	Carter Hayes

14.2 SPECIALTY CONSULTANTS

Cost Estimating:

Dharam Consulting

David Pearson
Robyn Glisson
Zack Everett
Jerry Jarnagin

Civil:

Dewberry

Sarah Purcell

Environmental Services:

Froehling & Robertson

Jesse Phillips